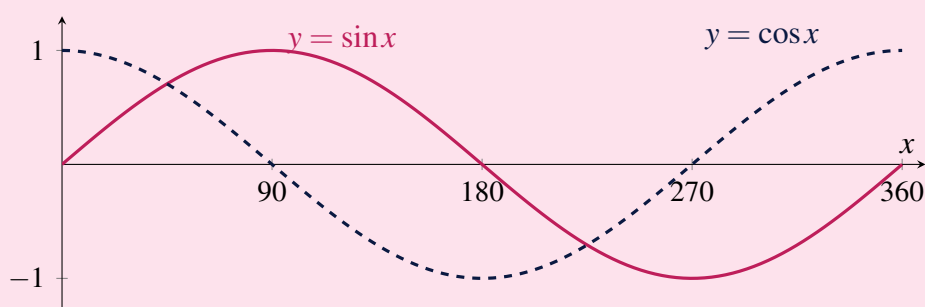


Advanced Trigonometry

Notes · Questions · Worked Answers

Pythagoras & right-angled trig · Exact values · Trig graphs & equations
($0-180^\circ$, $0-360^\circ$) · Area $\frac{1}{2}ab \sin C$ · Arcs, sectors & segments · Sine rule & the
ambiguous case · Cosine rule · Bearings · 3D trigonometry



How to use this booklet

- Read the pink **Key Notes** pages first.
- The questions are grouped by type into 13 sections, in syllabus order.
- Each white **Question** page is followed straight away by a yellow **Answer** page with full working — attempt every question before turning over.
- Work in the space under each question. Calculator allowed unless stated; give answers to 3 significant figures and angles to 1 decimal place.

Key Notes

0. Warm-up: Pythagoras and SOH CAH TOA

- Triangle = $\frac{1}{2} \times b \times h$ (h perpendicular to b)
- Pythagoras: $a^2 + b^2 = c^2$ (c = hypotenuse)
- 3D: diagonal of cuboid = $\sqrt{a^2 + b^2 + c^2}$

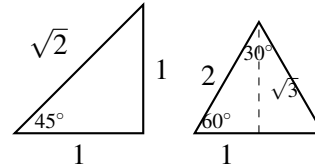
$$\sin \theta = \frac{\text{opp}}{\text{hyp}}, \quad \cos \theta = \frac{\text{adj}}{\text{hyp}}, \quad \tan \theta = \frac{\text{opp}}{\text{adj}}$$

Shortest distance from a point to a line is the **perpendicular** distance.

Angle of elevation / depression is measured from the horizontal.

1. Exact values (no calculator) — syllabus: PPT

θ	0°	30°	45°	60°	90°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	undefined



Memory trick: sin of 0, 30, 45, 60, 90 is $\frac{\sqrt{0}}{2}, \frac{\sqrt{1}}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, \frac{\sqrt{4}}{2}$; cos is the same list backwards; $\tan = \frac{\sin}{\cos}$.

2. Trig graphs and solving equations ($0-180^\circ$ and $0-360^\circ$)

- $y = \sin x, y = \cos x$: period 360° , values between -1 and 1 .
- $y = \tan x$: period 180° , asymptotes at $x = 90^\circ, 270^\circ$.
- Symmetry rules** (find the calculator value first):
 $\sin x$: $x, 180 - x$
 $\cos x$: $x, 360 - x$
 $\tan x$: $x, x + 180$
- $\sin x < 0 \Rightarrow$ answers in $180-360$. $\cos x < 0 \Rightarrow$ answers in $90-270$.

Example: $\sin x = -0.4, 0 \leq x \leq 360$.

Calc: $\sin^{-1}(0.4) = 23.6^\circ$ (reference angle)

$$\Rightarrow x = 180 + 23.6 = 203.6^\circ$$

$$x = 360 - 23.6 = 336.4^\circ$$

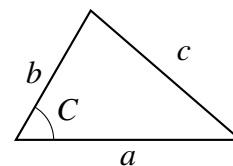
CAST: in quadrant 1 **A**ll positive, 2 **S**in, 3 **T**an, 4 **C**os.

For $0-180$ only: sin gives **two** answers (if $0 < k < 1$), cos and tan give **one**.

3. Area of a triangle

$$\text{Area} = \frac{1}{2} ab \sin C \quad (\text{two sides and the angle between them})$$

- Parallelogram = $ab \sin \theta$ (rhombus: $a^2 \sin \theta$).
- Regular n -gon in circle radius r : $n \times \frac{1}{2} r^2 \sin\left(\frac{360}{n}\right)$.
- Quadrilateral: split along a diagonal into two triangles.
- Reverse: $\sin C = \frac{2 \times \text{Area}}{ab}$ — remember C may be acute **or** $180 - C$.



4. Circles, sectors and segments

For angle θ at the centre, radius r :

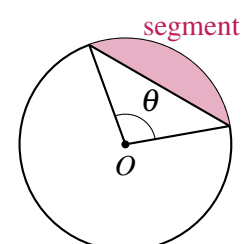
$$\text{Arc length} = \frac{\theta}{360} \times 2\pi r \quad \text{Sector area} = \frac{\theta}{360} \times \pi r^2$$

$$\text{Segment area} = \text{sector area} - \frac{1}{2} r^2 \sin \theta$$

$$\text{Chord length} = 2r \sin\left(\frac{\theta}{2}\right) \quad (\text{or use cosine rule})$$

$$\text{Segment perimeter} = \text{arc} + \text{chord}$$

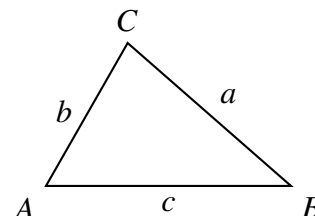
$$\text{Major segment} = \pi r^2 - \text{minor segment}$$



5. Sine rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} \quad \text{or} \quad \frac{\sin A}{a} = \frac{\sin B}{b}$$

- Use when you know a **side and its opposite angle** (a matching pair) plus one more side or angle.
- Finding a side: put sides on top. Finding an angle: put angles on top.
- Angles in a triangle add to 180° — use it to find the third angle first.



6. Sine rule — the ambiguous case

When finding an **angle** with the sine rule, $\sin^{-1}$ gives one acute answer θ , but $180^\circ - \theta$ has the **same sine**.

- Given angle A and sides a, b (angle not between them): if $a < b$ there may be **two** triangles.
- Check: is $180 - \theta + \text{the given angle} < 180^\circ$? If yes, both are possible.
- If $\sin \theta > 1 \Rightarrow$ **no triangle** exists. If $\sin \theta = 1 \Rightarrow$ exactly one (right-angled) triangle.
- If the question says “obtuse angle”, give $180 - \theta$.

7. Cosine rule

$$a^2 = b^2 + c^2 - 2bc \cos A \quad \cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

- Side: two sides & the **included** angle (SAS).
- Angle: all three sides (SSS). Largest angle is opposite the longest side.
- Negative $\cos A \Rightarrow A$ is obtuse (the cosine rule is **never** ambiguous).

Which rule?

Matching pair (side + opposite angle)? $\rightarrow$ **Sine rule**.

SAS or SSS? $\rightarrow$ **Cosine rule**.

Right angle? $\rightarrow$ SOH CAH TOA / Pythagoras.

8. Mixed problems: bearings and 3D

Bearings: measured **clockwise from North**, 3 figures.

Back bearing = bearing $\pm 180^\circ$.

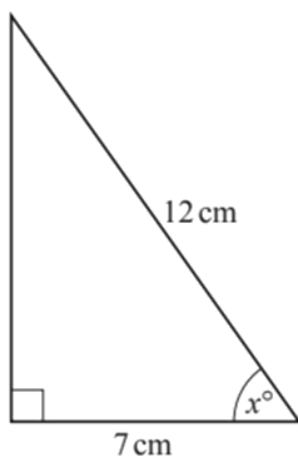
Draw North lines at every point; use co-interior angles (add to 180°) between parallel North lines.

3D: find a right-angled triangle that contains the line you need.

Cuboid diagonal = $\sqrt{a^2 + b^2 + c^2}$.

Angle between a line and a plane: drop a perpendicular to the plane, use the line, its projection and the height.

1

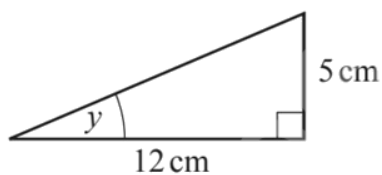


NOT TO
SCALE

Calculate the value of x .

$x = \dots\dots\dots$ [2]

2 Find, as a fraction, the value of $\sin y$.



NOT TO
SCALE

$\sin y = \dots\dots\dots$ [3]

3 The lengths of the sides of a triangle are 3 cm, 4 cm and 5 cm.

Find the sine of the smallest angle.

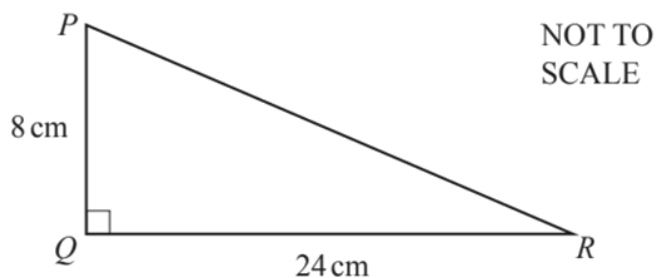
$\dots\dots\dots$ [1]

$$1 \quad \cos x = \frac{7}{12} \Rightarrow x = \cos^{-1}(0.5833) = \boxed{54.3^\circ}$$

$$2 \quad \text{Hypotenuse} = \sqrt{12^2 + 5^2} = 13. \quad \sin y = \frac{\text{opp}}{\text{hyp}} = \boxed{\frac{5}{13}} \Rightarrow y = \sin^{-1}\left(\frac{5}{13}\right) = \boxed{22.6^\circ}$$

$$3 \quad 3^2 + 4^2 = 5^2, \text{ so the triangle is right-angled; the smallest angle is opposite 3: } \sin = \boxed{\frac{3}{5}} \Rightarrow \text{angle} = \sin^{-1}(0.6) = \boxed{36.9^\circ}$$

4



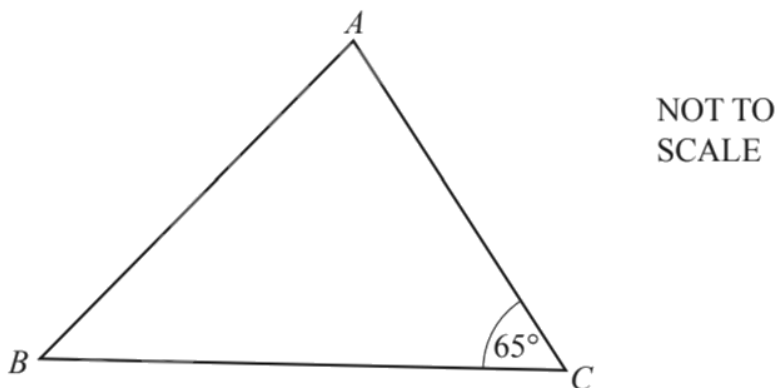
- (i) Calculate the area of triangle PQR .

..... cm^2 [2]

- (ii) Calculate angle PRQ .

Angle $PRQ =$ [2]

5



The shortest distance from B to AC is 12.8 cm .

Calculate BC .

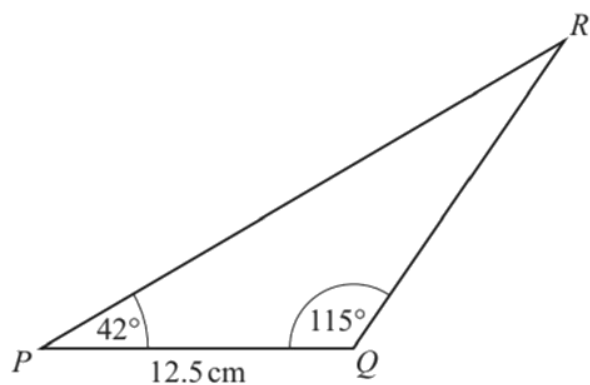
$BC =$ cm [3]

4 (i) $\frac{1}{2} \times 8 \times 24 = \boxed{96 \text{ cm}^2}$ (ii) $\tan PRQ = \frac{8}{24} \Rightarrow \boxed{18.4^\circ}$

5 The perpendicular from B to AC makes a right-angled triangle with hypotenuse BC :

$$\sin 65^\circ = \frac{12.8}{BC} \Rightarrow BC = \frac{12.8}{\sin 65^\circ} = \boxed{14.1 \text{ cm}}$$

6

NOT TO
SCALE

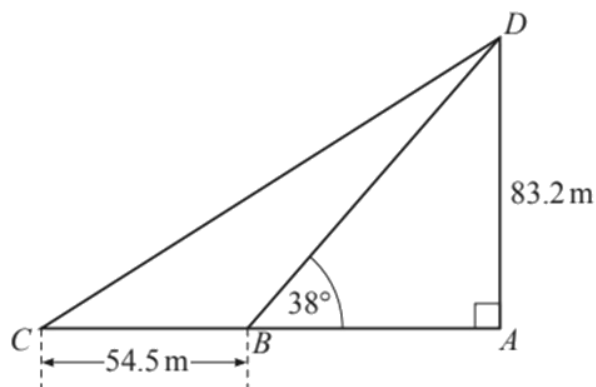
The diagram shows triangle PQR .

Calculate the shortest distance from Q to PR .

..... cm [3]

- 6** Shortest distance is the perpendicular from Q to PR : $d = PQ \sin 42^\circ = 12.5 \sin 42^\circ =$
8.36 cm

7

NOT TO
SCALE

ACD is a right-angled triangle.

B is on AC and $BC = 54.5$ m.

$AD = 83.2$ m and angle $ABD = 38^\circ$.

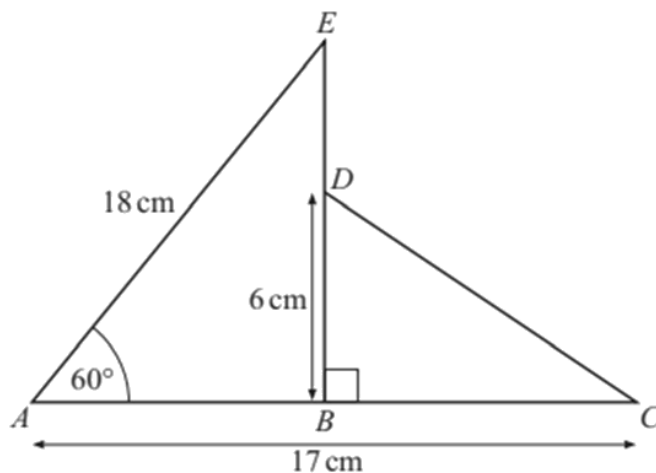
Calculate angle ACD .

Angle $ACD = \dots\dots\dots$ [5]

7 In triangle ABD : $AB = \frac{83.2}{\tan 38^\circ} = 106.49$ m. $AC = 106.49 + 54.5 = 160.99$ m.

$$\tan ACD = \frac{83.2}{160.99} \Rightarrow \boxed{\angle ACD = 27.3^\circ}$$

8

NOT TO
SCALE

The quadrilateral $ACDE$ is formed by two right-angled triangles ABE and BCD .
 $AC = 17$ cm, $AE = 18$ cm and $BD = 6$ cm.

(a) Show that $CD = 10$ cm.

[5]

(b) Find the perimeter of the quadrilateral $ACDE$.
 Give your answer in the form $p + k\sqrt{q}$.

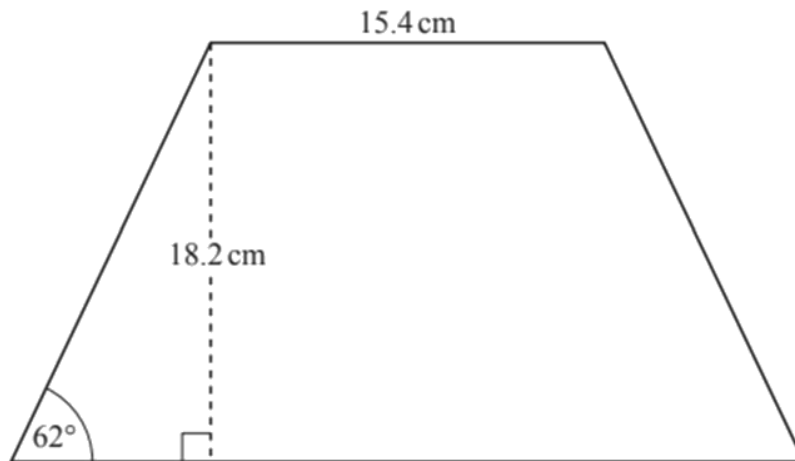
..... cm [4]

8 (a) $AB = 18 \cos 60^\circ = 9$, so $BC = 17 - 9 = 8$. $CD = \sqrt{8^2 + 6^2} = \sqrt{100} = 10$ ✓

(b) $BE = 18 \sin 60^\circ = 9\sqrt{3}$, so $DE = 9\sqrt{3} - 6$.

$$\text{Perimeter} = AC + CD + DE + EA = 17 + 10 + (9\sqrt{3} - 6) + 18 = \boxed{39 + 9\sqrt{3} \text{ cm}}$$

9

NOT TO
SCALE

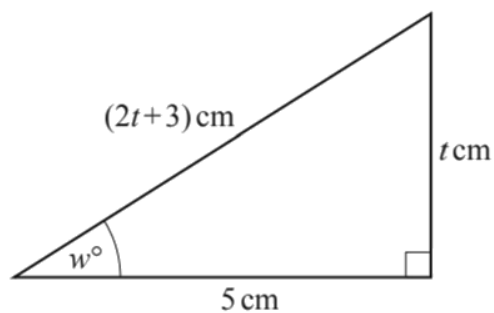
The diagram shows a trapezium.
The trapezium has one line of symmetry.

Work out the area of the trapezium.

..... cm^2 [4]

9 Overhang each side = $\frac{18.2}{\tan 62^\circ} = 9.677$; bottom = $15.4 + 2(9.677) = 34.75$.
Area = $\frac{1}{2}(15.4 + 34.75) \times 18.2 = \boxed{456 \text{ cm}^2}$

10

NOT TO
SCALE

The diagram shows a right-angled triangle.

Find the value of w .

$w = \dots\dots\dots$ [7]

10 Pythagoras: $(2t + 3)^2 = t^2 + 5^2 \Rightarrow 4t^2 + 12t + 9 = t^2 + 25 \Rightarrow 3t^2 + 12t - 16 = 0$
 $t = \frac{-12 + \sqrt{144 + 192}}{6} = 1.055$ (positive root). $\tan w = \frac{t}{5} = 0.2110 \Rightarrow \boxed{w = 11.9}$

11 Complete the table using exact values. Do not use a calculator.

θ	0°	30°	45°	60°	90°
$\sin \theta$					
$\cos \theta$					
$\tan \theta$					

[5]

- 12
Without using a calculator, find the exact value of
- (a) $\sin 30^\circ + \cos 60^\circ$
(b) $\tan 45^\circ \times \sin 60^\circ$
(c) $\sin^2 45^\circ + \cos^2 30^\circ$

(d) $\frac{\tan 60^\circ}{\cos 30^\circ}$
(e) $2 \sin 60^\circ \cos 60^\circ$

(a) (b) (c) (d) (e) [5]

13 Find the acute angle x in each case.

- (a) $\sin x = \frac{\sqrt{2}}{2}$
(b) $\tan x = \frac{1}{\sqrt{3}}$
(c) $2 \cos x = 1$

[3]

11

θ	0°	30°	45°	60°	90°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	undefined

12 (a) $\frac{1}{2} + \frac{1}{2} = \boxed{1}$ (b) $1 \times \frac{\sqrt{3}}{2} = \boxed{\frac{\sqrt{3}}{2}}$ (c) $\frac{2}{4} + \frac{3}{4} = \boxed{\frac{5}{4}}$ (d) $\sqrt{3} \div \frac{\sqrt{3}}{2} = \boxed{2}$ (e)

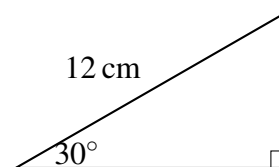
$2 \times \frac{\sqrt{3}}{2} \times \frac{1}{2} = \boxed{\frac{\sqrt{3}}{2}}$

13 (a) $\boxed{45^\circ}$ (b) $\frac{1}{\sqrt{3}} = \tan 30^\circ \Rightarrow \boxed{30^\circ}$ (c) $\cos x = \frac{1}{2} \Rightarrow \boxed{60^\circ}$

- 14** (a) A right-angled triangle has hypotenuse 12 cm and one angle of 30° .

Find the exact lengths of the other two sides.

- (b) Another right-angled triangle has an angle of 60° whose adjacent side is 5 cm. Find the exact length of the opposite side and of the hypotenuse.



NOT TO SCALE

- 15** An equilateral triangle has sides of length 6 cm.

- (a) Show that its height is exactly $3\sqrt{3}$ cm.

- (b) Find its exact area.

[4]

cm^2 [4]

- 16** A triangle has sides 8 cm and 10 cm with an included angle of 60° .

Find the exact area of the triangle, giving your answer in the form $k\sqrt{3}$.

cm^2 [3]

14 (a) opposite = $12 \sin 30^\circ = \boxed{6 \text{ cm}}$; adjacent = $12 \cos 30^\circ = \boxed{6\sqrt{3} \text{ cm}}$

(b) opposite = $5 \tan 60^\circ = \boxed{5\sqrt{3} \text{ cm}}$; hypotenuse = $\frac{5}{\cos 60^\circ} = \boxed{10 \text{ cm}}$

15 (a) Half-triangle: hypotenuse 6, base 3; height = $\sqrt{36 - 9} = \sqrt{27} = 3\sqrt{3} \checkmark$

(b) $\frac{1}{2} \times 6 \times 3\sqrt{3} = \boxed{9\sqrt{3} \text{ cm}^2}$

16 $\frac{1}{2} \times 8 \times 10 \times \sin 60^\circ = 40 \times \frac{\sqrt{3}}{2} = \boxed{20\sqrt{3} \text{ cm}^2}$

- 17 A right-angled isosceles triangle has shorter sides of length 7 cm.

Find the exact length of the hypotenuse and use the triangle to show that $\cos 45^\circ = \frac{\sqrt{2}}{2}$.

cm [2]

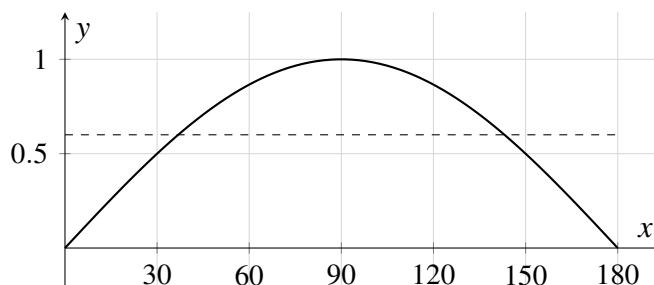
17 Hypotenuse = $\sqrt{49+49} = \boxed{7\sqrt{2} \text{ cm}}$; $\cos 45^\circ = \frac{7}{7\sqrt{2}} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2} \checkmark$

- 18** The graph of $y = \sin x$ for $0^\circ \leq x \leq 180^\circ$ is shown, with the line $y = 0.6$.

(a) How many solutions does $\sin x = 0.6$ have in this range?

(b) One solution is $x = 36.9^\circ$.

Use the symmetry of the graph to find the other.



(a) (b) $x =$ [3]

- 19** Solve each equation for $0^\circ \leq x \leq 180^\circ$. Give your answers correct to 1 decimal place.

(a) $\sin x = 0.35$

(b) $\cos x = -0.42$

(c) $\tan x = -2.5$

(d) $4 \sin x - 1 = 2$

(e) $5 \cos x + 2 = 0$

(f) $3 \tan x = 7$

[10]

- 20** Write down the coordinates of the maximum point of $y = \sin x$ and of the minimum point of $y = \cos x$ for $0^\circ \leq x \leq 180^\circ$.

max (...,...) min [2]

18 (a) The line cuts the curve twice: $\boxed{2}$. (b) Symmetry about $x = 90^\circ$: $180 - 36.9 = \boxed{143.1^\circ}$

19 (a) $\sin^{-1} 0.35 = 20.5^\circ$; $180 - 20.5$: $\boxed{20.5^\circ, 159.5^\circ}$ (b) $\cos^{-1}(-0.42) = \boxed{114.8^\circ}$
(c) $\tan^{-1}(-2.5) = -68.2^\circ$; $+180$: $\boxed{111.8^\circ}$ (d) $\sin x = 0.75$: $\boxed{48.6^\circ, 131.4^\circ}$
(e) $\cos x = -0.4$: $\boxed{113.6^\circ}$ (f) $\tan x = \frac{7}{3}$: $\boxed{66.8^\circ}$ (next is 246.8° , out of range)

20 Max of $\sin x$: $\boxed{(90^\circ, 1)}$; min of $\cos x$: $\boxed{(180^\circ, -1)}$

- 21** (a) Given that $\sin 52^\circ = 0.788$, find another angle between 0° and 180° whose sine is 0.788.
 (b) Given that $\cos 70^\circ = 0.342$, find the angle between 0° and 180° whose cosine is -0.342 .

(a) (b) [2]

- 22** For $0^\circ \leq x \leq 180^\circ$, state the number of solutions of

(a) $\sin x = 0.4$ (b) $\sin x = 1$ (c) $\sin x = -0.2$ (d) $\cos x = -0.9$

(a) (b) (c) (d) [4]

- 23** Solve $\sin^2 x = 0.25$ for $0^\circ \leq x \leq 180^\circ$.

$x =$ [3]

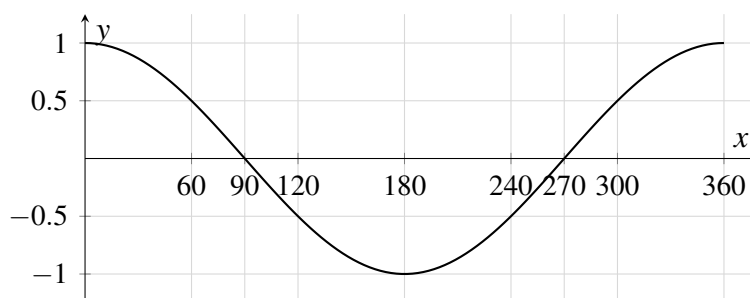
21 (a) $180 - 52 = \boxed{128^\circ}$ (b) $\cos(180 - 70) = -\cos 70$: $\boxed{110^\circ}$

22 (a) $\boxed{2}$ (b) $\boxed{1}$ ($x = 90^\circ$) (c) $\boxed{0}$ ($\sin x \geq 0$ here) (d) $\boxed{1}$

23 $\sin x = \pm 0.5$; in $0-180$ only $\sin x = 0.5$ works: $\boxed{x = 30^\circ, 150^\circ}$

- 24** The graph of $y = \cos x$ for $0^\circ \leq x \leq 360^\circ$ is shown.

Use the graph to solve the equation $\cos x = 0.5$.



$x = \dots\dots\dots$ [2]

- 25** Solve for $0^\circ \leq x \leq 360^\circ$, giving answers to 1 decimal place where necessary.

(a) $\cos x = 0.3$

(b) $\tan x = 1.8$

(c) $\sin x = -0.72$

(d) $2 \cos x + 1 = 0$

(e) $3 \tan x + 4 = 0$

[10]

- 26** One solution of $\cos x = -0.8$ is $x = 143.1^\circ$.
Find the other solution for $0^\circ \leq x \leq 360^\circ$.

$x = \dots\dots\dots$ [1]

24 Line $y = 0.5$ meets the curve at $x = 60^\circ$ and 300°

25 (a) $72.5^\circ, 360 - 72.5$: $72.5^\circ, 287.5^\circ$ (b) $60.9^\circ, +180$: $60.9^\circ, 240.9^\circ$

(c) reference $\sin^{-1} 0.72 = 46.1^\circ$: $226.1^\circ, 313.9^\circ$ (d) $\cos x = -\frac{1}{2}$: $120^\circ, 240^\circ$

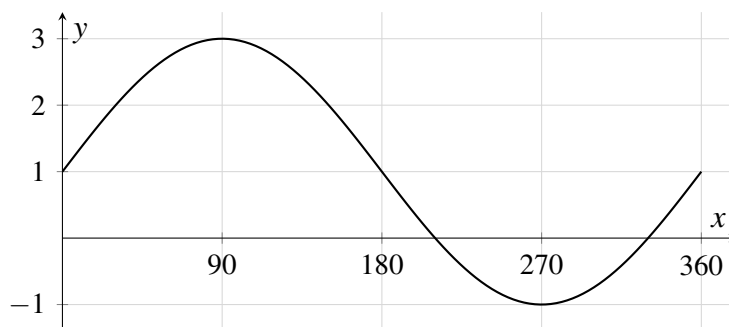
(e) $\tan x = -\frac{4}{3}$, calc -53.1° : $126.9^\circ, 306.9^\circ$

26 $360 - 143.1 = 216.9^\circ$

- 27** The graph of $y = 2 \sin x + 1$ for $0^\circ \leq x \leq 360^\circ$ is shown.

(a) Write down the maximum and minimum values of y and where they occur.

(b) Solve $2 \sin x + 1 = 0$.



(b) $x = \dots\dots\dots$ [3]

- 28** (a) Write down the equations of the asymptotes of $y = \tan x$ for $0^\circ \leq x \leq 360^\circ$.

(b) State the period of $y = \tan x$ and the period of $y = \sin x$.

[3]

- 29** Solve $\tan^2 x = 3$ for $0^\circ \leq x \leq 360^\circ$.

$x = \dots\dots\dots$ [3]

- 30** Solve $\sin x = \cos x$ for $0^\circ \leq x \leq 360^\circ$.

$x = \dots\dots\dots$ [2]

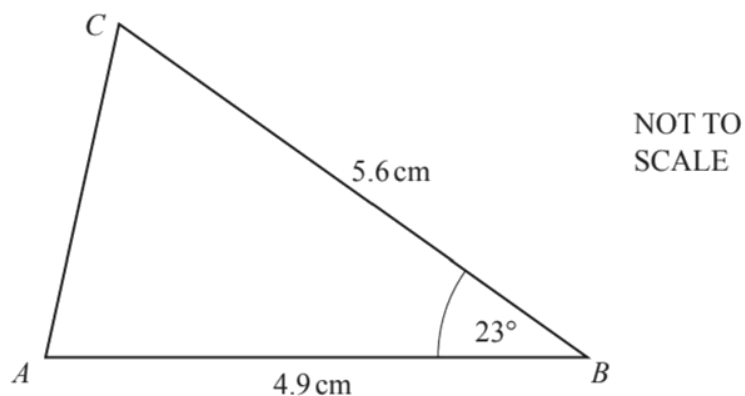
27 (a) Max $\boxed{3}$ at $x = 90^\circ$; min $\boxed{-1}$ at $x = 270^\circ$. (b) $\sin x = -\frac{1}{2}$: $\boxed{210^\circ, 330^\circ}$

28 (a) $\boxed{x = 90^\circ, x = 270^\circ}$ (b) $\tan$: $\boxed{180^\circ}$; $\sin$: $\boxed{360^\circ}$

29 $\tan x = \pm\sqrt{3}$: $\boxed{60^\circ, 120^\circ, 240^\circ, 300^\circ}$

30 Divide by $\cos x$: $\tan x = 1 \Rightarrow \boxed{45^\circ, 225^\circ}$

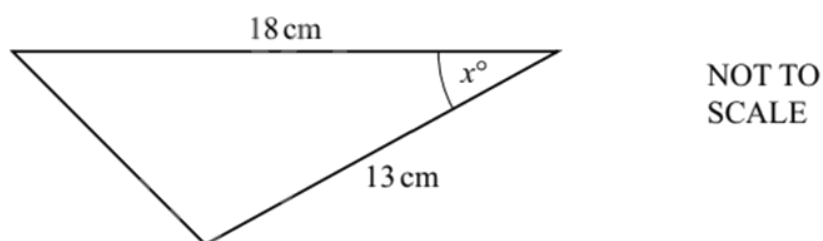
31



Calculate the area of triangle ABC .

..... cm^2 [2]

32



The area of the triangle is 50 cm^2 .

Calculate the value of $\sin x$.

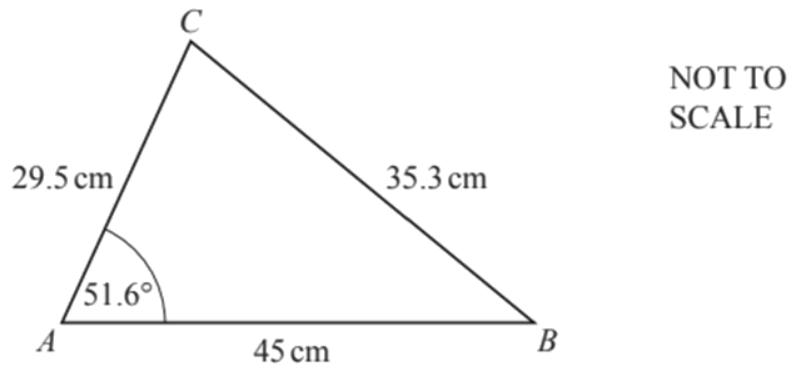
$\sin x =$ [2]

31 Angle 23° is between 5.6 and 4.9: $\frac{1}{2} \times 4.9 \times 5.6 \times \sin 23^\circ = \boxed{5.36 \text{ cm}^2}$

32 $50 = \frac{1}{2} \times 18 \times 13 \times \sin x = 117 \sin x \Rightarrow \sin x = \frac{50}{117} = \boxed{0.427}$

In degrees: $x = \sin^{-1}(0.4274) = \boxed{25.3^\circ}$

33



In triangle ABC , $AB = 45 \text{ cm}$, $AC = 29.5 \text{ cm}$, $BC = 35.3 \text{ cm}$ and angle $CAB = 51.6^\circ$.

- (i) Calculate angle ABC .

Angle $ABC = \dots\dots\dots$ [3]

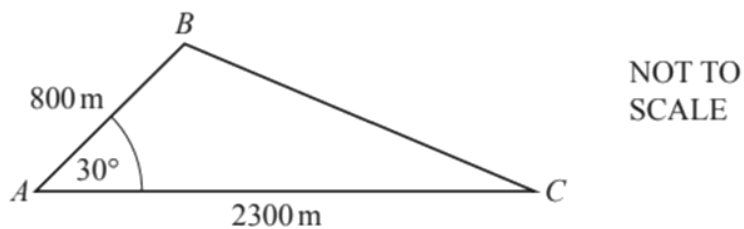
- (ii) Calculate the area of triangle ABC .

$\dots\dots\dots \text{ cm}^2$ [2]

33 (i) Sine rule: $\frac{\sin B}{29.5} = \frac{\sin 51.6^\circ}{35.3} \Rightarrow \sin B = 0.6550 \Rightarrow \boxed{\angle ABC = 40.9^\circ}$

(ii) $\frac{1}{2} \times 45 \times 29.5 \times \sin 51.6^\circ = \boxed{520 \text{ cm}^2}$

34



The diagram shows some land in the shape of a triangle ABC .

Houses are built on this land.

Each house requires 400 m^2 of land.

Find the greatest number of houses that can be built on this land.

..... [3]

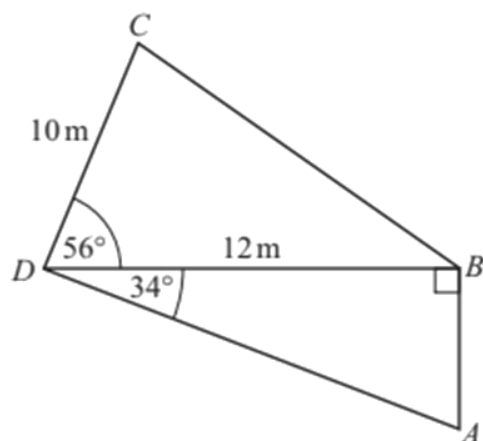
35 Find the area of a regular hexagon with side length 7.4 cm.

..... cm^2 [3]

34 $\text{Area} = \frac{1}{2} \times 800 \times 2300 \times \sin 30^\circ = 460\,000 \text{ m}^2$. $460\,000 \div 400 = \boxed{1150 \text{ houses}}$

35 6 equilateral triangles: $6 \times \frac{1}{2} \times 7.4^2 \times \sin 60^\circ = \boxed{142 \text{ cm}^2}$

36

NOT TO
SCALE

The diagram shows a quadrilateral $ABCD$.
 $CD = 10$ m and $DB = 12$ m.
 Angle $DBA = 90^\circ$, angle $CDB = 56^\circ$ and angle $ADB = 34^\circ$.

(a) Calculate the length of AB .

$AB = \dots\dots\dots$ m [2]

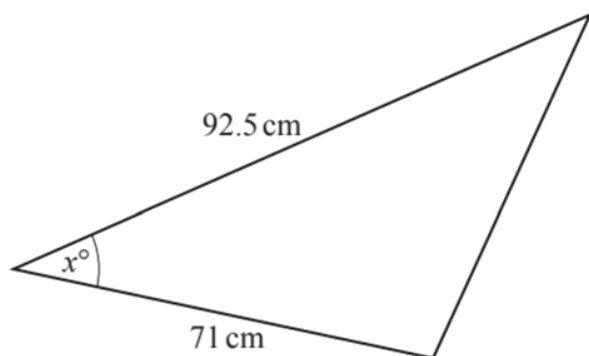
(b) Calculate the area of the quadrilateral $ABCD$.

$\dots\dots\dots$ m^2 [3]

36 (a) $\tan 34^\circ = \frac{AB}{12} \Rightarrow AB = 12 \tan 34^\circ = \boxed{8.09 \text{ m}}$

(b) $ABD = \frac{1}{2} \times 12 \times 8.094 = 48.56$; $BCD = \frac{1}{2} \times 10 \times 12 \times \sin 56^\circ = 49.74$. Total
= $\boxed{98.3 \text{ m}^2}$

37

NOT TO
SCALE

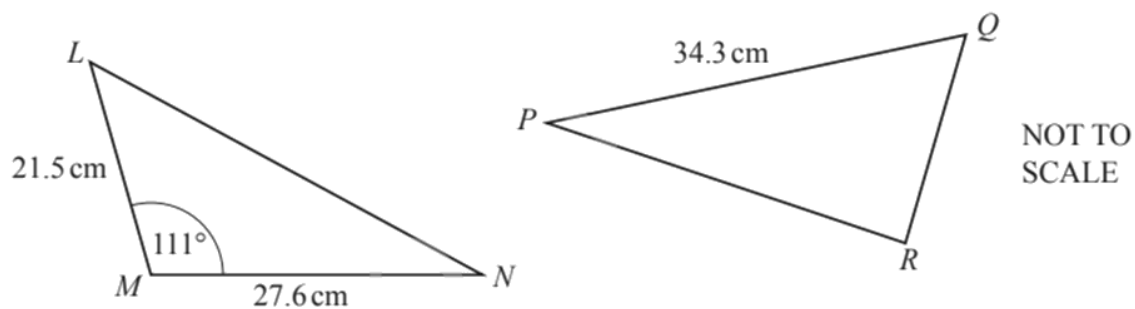
The diagram shows a triangle with an acute angle marked x° .
The area of the triangle is 2143 cm^2 .

Work out the value of x .

$x = \dots\dots\dots$ [2]

$$37 \quad 2143 = \frac{1}{2} \times 92.5 \times 71 \times \sin x \Rightarrow \sin x = \frac{4286}{6567.5} = 0.6526 \Rightarrow \boxed{x = 40.7}$$

38



Triangle PQR has the same area as triangle LMN .

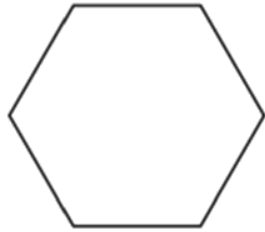
Calculate the shortest distance from R to the line PQ .

..... cm [3]

38 Area $LMN = \frac{1}{2} \times 21.5 \times 27.6 \times \sin 111^\circ = 277.0 \text{ cm}^2$.

Area $PQR = \frac{1}{2} \times PQ \times h \Rightarrow 277.0 = \frac{1}{2} \times 34.3 \times h \Rightarrow h = \boxed{16.2 \text{ cm}}$

39.

NOT TO
SCALE

The diagram shows a regular hexagon.
The area of the hexagon is 127.3 cm^2 .

- (i) Show that the length of one side of the hexagon is 7.0 cm , correct to 1 decimal place.

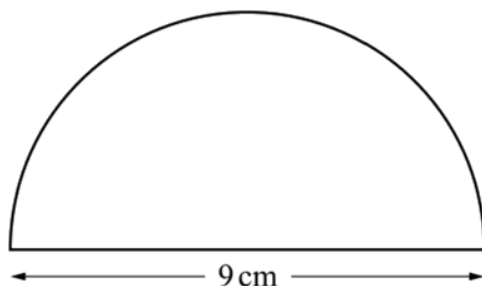
[4]

$$\mathbf{39} \quad 6 \times \frac{1}{2}s^2 \sin 60^\circ = 127.3 \Rightarrow s^2 = \frac{127.3}{3 \sin 60^\circ} = 48.997 \Rightarrow s = 6.9998 = 7.0 \text{ cm } \checkmark$$

- 40 Calculate the circumference of a circle with radius 4.7 cm.

..... cm [2]

41



NOT TO
SCALE

The diagram shows a semicircle with diameter 9 cm.

Calculate the total perimeter of this semicircle.
Give your answer in exact form.

..... cm [3]

- 42 The total perimeter of a semicircle is 19.02 cm.

Calculate the radius of the semicircle.

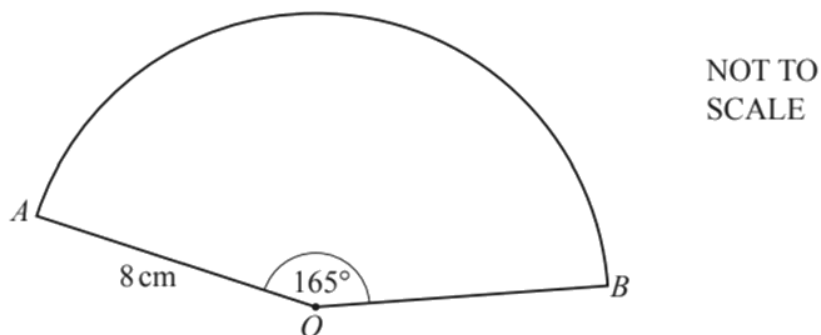
..... cm [3]

$$40 \quad 2\pi r = 2\pi \times 4.7 = \boxed{29.5 \text{ cm}}$$

$$41 \quad \text{Arc} = \frac{1}{2}\pi d = \frac{1}{2}\pi(9) = 4.5\pi; \text{ perimeter} = \boxed{4.5\pi + 9 \text{ cm}}$$

$$42 \quad \text{Perimeter} = \pi r + 2r = r(\pi + 2) = 19.02 \Rightarrow r = \frac{19.02}{\pi + 2} = \boxed{3.70 \text{ cm}}$$

43



The diagram shows a sector of a circle with centre O , radius 8 cm and sector angle 165° .

(a) Calculate the total perimeter of the sector.

..... cm [3]

44 The perimeter of a sector of a circle with radius 8 cm is 26 cm .

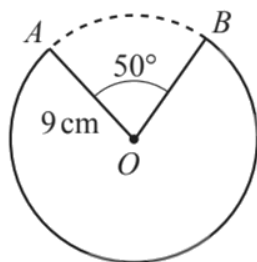
Calculate the angle of this sector.

..... [3]

43 $\text{Arc} = \frac{165}{360} \times 2\pi \times 8 = 23.04$; $\text{perimeter} = 23.04 + 8 + 8 = \boxed{39.0 \text{ cm}}$

44 $\text{Arc} = 26 - 8 - 8 = 10$. $\frac{\theta}{360} \times 2\pi \times 8 = 10 \Rightarrow \theta = \frac{3600}{16\pi} = \boxed{71.6^\circ}$

45

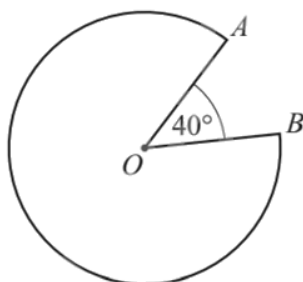
NOT TO
SCALE

The diagram shows a circle of radius 9 cm, centre O .
The minor sector AOB , with sector angle 50° , is removed from the circle.

Calculate the length of the major arc AB .

..... cm [3]

46

NOT TO
SCALE

The diagram shows a sector of a circle, centre O .
The radius of the circle is 6 cm.

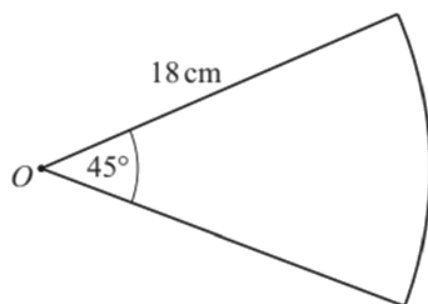
Calculate the length of the major arc AB .
Give your answer in its simplest form in terms of π .

.....cm [3]

45 Major angle = $360 - 50 = 310^\circ$. Arc = $\frac{310}{360} \times 2\pi \times 9 = \boxed{48.7 \text{ cm}}$

46 Major angle = 320° : $\frac{320}{360} \times 2\pi \times 6 = \boxed{\frac{32}{3}\pi \text{ cm}}$

47

NOT TO
SCALE

The diagram shows a sector of a circle, centre O .
The length of the arc is $n\pi\text{ cm}$.

Find the value of n .

$n = \dots\dots\dots$ [2]

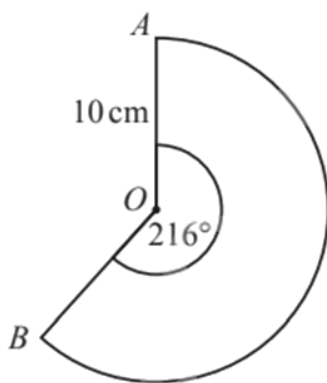
- 48 Calculate the area of the sector of a circle with radius 65 mm and sector angle 42° .
Give your answer in square centimetres.

$\dots\dots\dots\text{ cm}^2$ [3]

$$47 \quad \text{Arc} = \frac{45}{360} \times 2\pi \times 18 = 4.5\pi \Rightarrow \boxed{n = 4.5}$$

$$48 \quad r = 65 \text{ mm} = 6.5 \text{ cm}: \frac{42}{360} \times \pi \times 6.5^2 = \boxed{15.5 \text{ cm}^2}$$

49

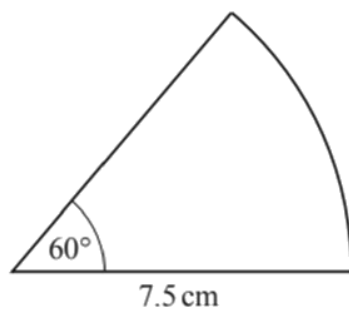

NOT TO
SCALE

AOB is a sector of a circle, centre O .
 $AO = 10$ cm and the sector angle is 216° .

- (i) Calculate the length of the arc of this sector.
 Give your answer as a multiple of π .

.....cm [2]

50


NOT TO
SCALE

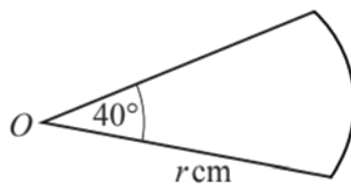
Calculate the area of this sector of a circle.

..... cm² [2]

$$49 \quad \frac{216}{360} \times 2\pi \times 10 = \boxed{12\pi \text{ cm}}$$

$$50 \quad \frac{60}{360} \times \pi \times 7.5^2 = \boxed{29.5 \text{ cm}^2}$$

51

NOT TO
SCALE

The diagram shows an arc of a circle, centre O , radius $r \text{ cm}$.
The length of the arc is $k\pi r \text{ cm}$.

Find the value of k .

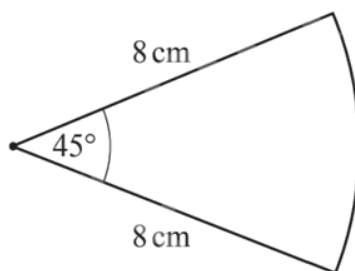
Give your answer as a fraction in its simplest form.

$k = \dots\dots\dots$ [2]

52

Find the area of the sector.

Give your answer, in terms of π , in its simplest form.

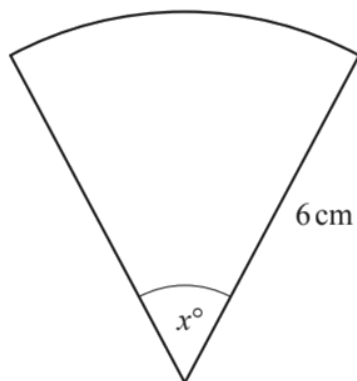
NOT TO
SCALE

$\dots\dots\dots \text{ cm}^2$ [2]

$$51 \quad \frac{40}{360} \times 2\pi r = \frac{2}{9}\pi r \Rightarrow \boxed{k = \frac{2}{9}}$$

$$52 \quad \frac{45}{360} \times \pi \times 8^2 = \boxed{8\pi \text{ cm}^2}$$

53

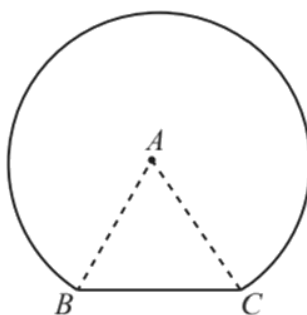

NOT TO
SCALE

The area of this sector is $5\pi \text{ cm}^2$.

Find the value of x .

$x =$ [3]

54


NOT TO
SCALE

The diagram shows a shape made from an equilateral triangle ABC and a sector of a circle. Points B and C lie on the circle, centre A .

The side length of the equilateral triangle is 12.4 cm.

Work out the perimeter of the shape.

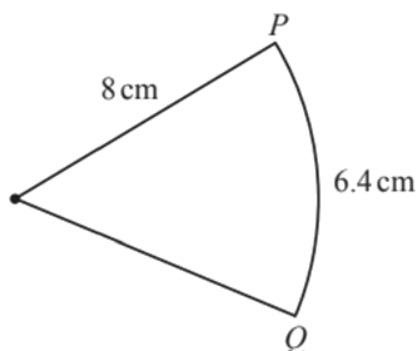
..... cm [3]

53 $\frac{x}{360} \times \pi \times 6^2 = 5\pi \Rightarrow \frac{x}{10} = 5 \Rightarrow \boxed{x = 50}$

54 Radius = 12.4, sector angle = $360 - 60 = 300^\circ$.

Arc = $\frac{300}{360} \times 2\pi \times 12.4 = 64.93$; perimeter = $64.93 + 12.4 = \boxed{77.3 \text{ cm}}$

55

NOT TO
SCALE

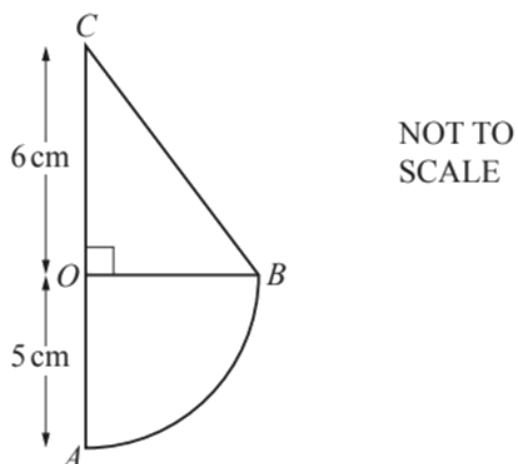
The diagram shows a sector of a circle of radius 8 cm.
The length of the arc PQ is 6.4 cm.

Find the area of the sector.

..... cm^2 [4]

55 $\frac{\theta}{360} \times 2\pi(8) = 6.4 \Rightarrow \frac{\theta}{360} = 0.1273$. Area = $0.1273 \times \pi \times 64 = \boxed{25.6 \text{ cm}^2}$ (or $\frac{1}{2}r \times \text{arc} = \frac{1}{2} \times 8 \times 6.4$)

56



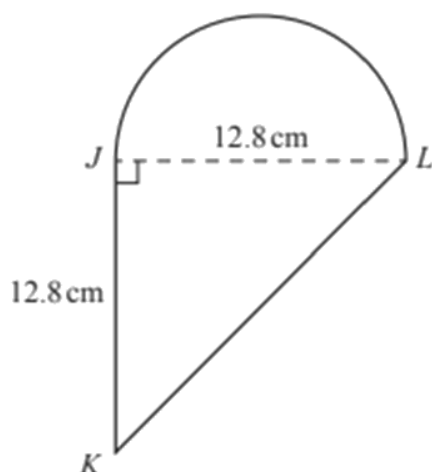
The diagram shows a shape made from a quarter-circle, OAB , and a right-angled triangle OBC . The radius of the circle is 5 cm and $OC = 6$ cm.

Calculate the area of the shape.

..... cm^2 [3]

56 Quarter circle $= \frac{1}{4}\pi(5)^2 = 19.63$; triangle $OBC = \frac{1}{2} \times 5 \times 6 = 15$. Total = 34.6 cm^2

57

NOT TO
SCALE

The diagram shows a shape made from a triangle JKL and a semicircle with diameter JL . JKL is an isosceles right-angled triangle with $JK = JL = 12.8$ cm.

(a) Calculate the area of this shape.

..... cm^2 [3]

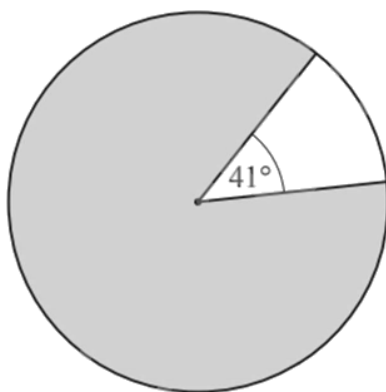
(b) Calculate the perimeter of this shape.

..... cm [4]

57 (a) Triangle = $\frac{1}{2} \times 12.8^2 = 81.92$; semicircle = $\frac{1}{2}\pi(6.4)^2 = 64.34$. Total = $\boxed{146 \text{ cm}^2}$

(b) $KL = \sqrt{12.8^2 + 12.8^2} = 18.10$; arc = $\pi \times 6.4 = 20.11$. Perimeter = $12.8 + 18.10 + 20.11 = \boxed{51.0 \text{ cm}}$

58

NOT TO
SCALE

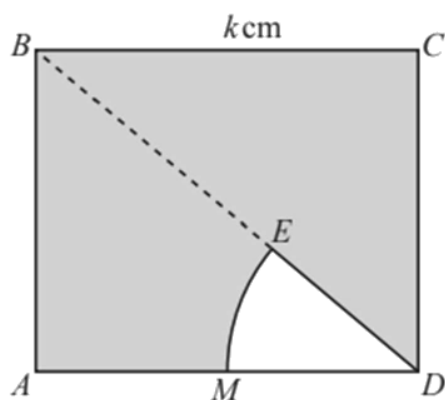
The diagram shows two sectors of a circle.
The major sector is shaded.
The area of the major sector is 74.5 cm^2 .

Calculate the radius of the circle.

..... cm [3]

58 Major angle = 319° : $\frac{319}{360}\pi r^2 = 74.5 \Rightarrow r^2 = 26.76 \Rightarrow \boxed{r = 5.17 \text{ cm}}$

59

NOT TO
SCALE

The diagram shows a square $ABCD$ with side length k cm.
 MDE is a sector of a circle, centre D .
 E lies on the diagonal, BD , of the square.
 M is the midpoint of AD .

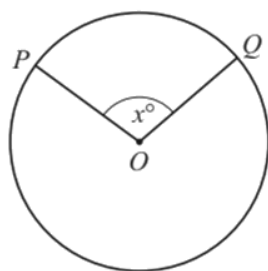
Find the percentage of the square that is shaded.

..... % [4]

59 Sector radius = $\frac{k}{2}$, angle 45° : area = $\frac{45}{360}\pi\left(\frac{k}{2}\right)^2 = \frac{\pi k^2}{32}$.

Shaded = $k^2 - \frac{\pi k^2}{32}$; percentage = $\left(1 - \frac{\pi}{32}\right) \times 100 = \boxed{90.2\%}$

60

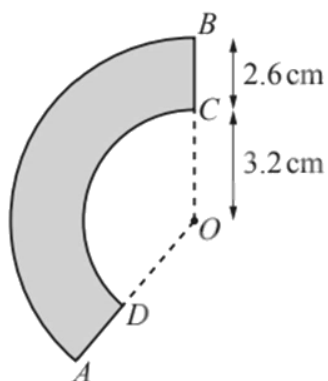
NOT TO
SCALE

In the circle, centre O , the length of the minor arc PQ is $\frac{3}{7}$ of the length of the major arc PQ .
Show that $x = 108$.

[3]

60 $\frac{x}{360} \times 2\pi r = \frac{3}{7} \times \frac{360-x}{360} \times 2\pi r \Rightarrow 7x = 3(360-x) \Rightarrow 10x = 1080 \Rightarrow x = 108 \checkmark$

61

NOT TO
SCALE

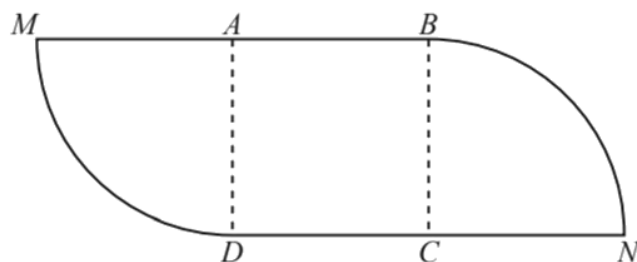
The diagram shows a shape, $ABCD$, formed by the sectors of two circles with the same centre O . Both sector angles are 140° , $OC = 3.2$ cm and $CB = 2.6$ cm. The area of the shape is $k\pi$ cm².

Find the value of k .

$k = \dots\dots\dots$ [3]

61 $OB = 3.2 + 2.6 = 5.8$. $\text{Area} = \frac{140}{360}\pi(5.8^2 - 3.2^2) = \frac{7}{18}\pi(23.4) = 9.1\pi \Rightarrow \boxed{k = 9.1}$

62

NOT TO
SCALE

The diagram shows a shape made from a square $ABCD$ and two equal sectors of a circle.
The square has side 11 cm.
 MAB and DCN are straight lines.

- (i) Calculate the area of the shape.

..... cm^2 [3]

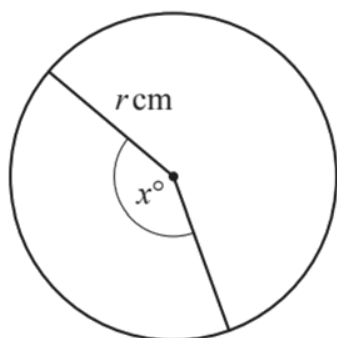
- (ii) Calculate the perimeter of the shape.

..... cm [3]

62 (i) Square = 121; two quarter circles radius 11 = $2 \times \frac{1}{4}\pi(11)^2 = 60.5\pi$. Total = $\boxed{311 \text{ cm}^2}$

(ii) $MB = 22$, $DN = 22$, two quarter arcs = $2 \times \frac{1}{4} \times 2\pi(11) = 11\pi$. Perimeter = $44 + 11\pi = \boxed{78.6 \text{ cm}}$

63

NOT TO
SCALE

The diagram shows a circle, radius $r \text{ cm}$ and minor sector angle x° .
The **perimeter** of the major sector is three times the **perimeter** of the minor sector.

Show that $x = \frac{90(\pi - 2)}{\pi}$.

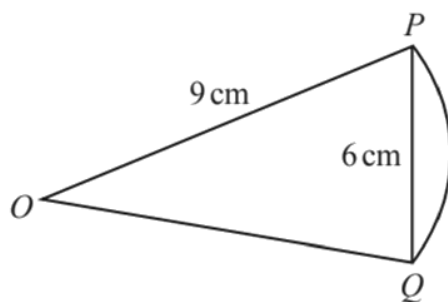
[4]

63 $2r + \frac{360-x}{360} 2\pi r = 3 \left(2r + \frac{x}{360} 2\pi r \right)$. Divide by r , multiply by 360:

$$720 + 2\pi(360 - x) = 2160 + 6\pi x \Rightarrow 720\pi - 1440 = 8\pi x \Rightarrow x = \frac{90\pi - 180}{\pi} = \frac{90(\pi - 2)}{\pi}$$

✓

64

NOT TO
SCALE

The diagram shows a sector of a circle with centre O and radius 9 cm .
The length of the chord PQ is 6 cm .

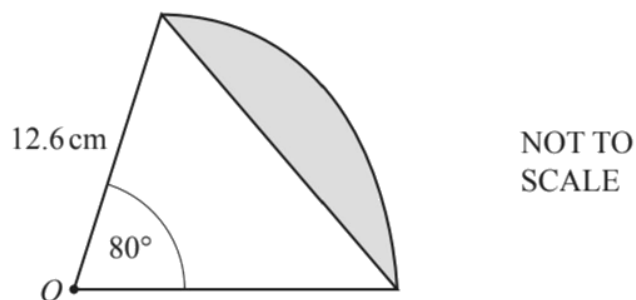
Calculate the length of the arc PQ .

..... cm [3]

64 Half the chord: $\sin \frac{\theta}{2} = \frac{3}{9} \Rightarrow \frac{\theta}{2} = 19.47^\circ \Rightarrow \theta = 38.94^\circ$

$$\text{Arc} = \frac{38.94}{360} \times 2\pi \times 9 = \boxed{6.12 \text{ cm}}$$

65



The diagram shows a sector of a circle, centre O , radius 12.6 cm .

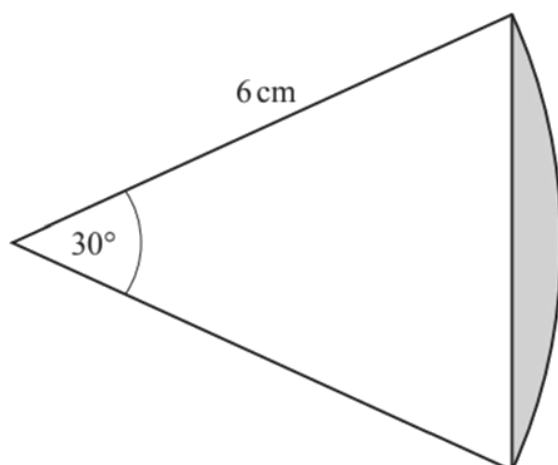
Calculate the perimeter of the shaded segment.

..... cm [4]

65 Arc = $\frac{80}{360} \times 2\pi \times 12.6 = 17.59$; chord² = $2(12.6)^2 - 2(12.6)^2 \cos 80^\circ \Rightarrow$ chord = 16.20

Perimeter = 33.8 cm

66



The area of the shaded segment is $(a\pi + b)\text{ cm}^2$.

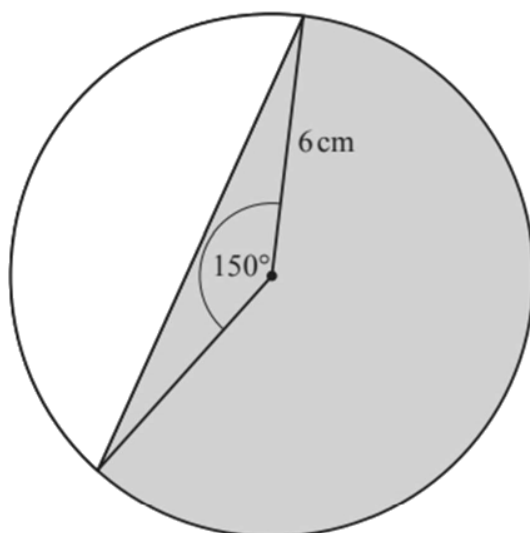
Find the value of a and the value of b .

$a =$

$b =$ [4]

66 Sector = $\frac{30}{360}\pi(36) = 3\pi$; triangle = $\frac{1}{2}(6)(6)\sin 30^\circ = 9$. Segment = $3\pi - 9 \Rightarrow$
 $a = 3, b = -9$

67

NOT TO
SCALE

A sector of a circle with radius 6 cm has a sector angle of 150° .

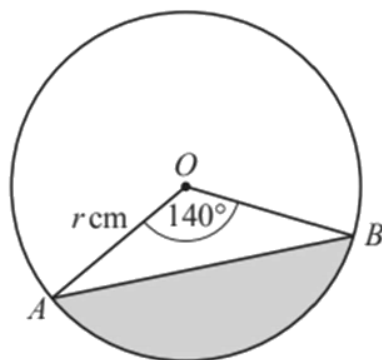
Find the exact value of the area of the shaded region.
Give your answer in its simplest form.

..... cm^2 [4]

67 Shaded = major sector (210°) + triangle.

$$\text{Sector} = \frac{210}{360}\pi(6)^2 = 21\pi; \text{ triangle} = \frac{1}{2} \times 6 \times 6 \times \sin 150^\circ = 9. \text{ Shaded} = \boxed{21\pi + 9 \text{ cm}^2}$$

68

NOT TO
SCALE

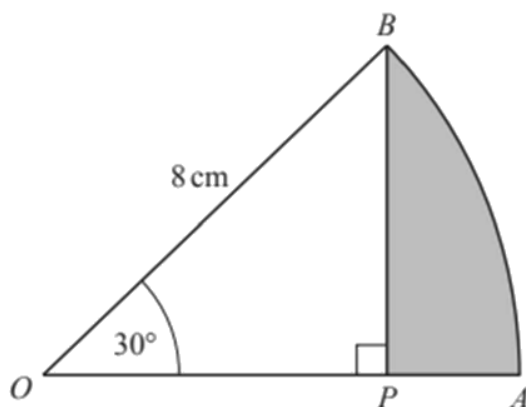
A and B are points on a circle, centre O , radius r cm.
The area of the shaded segment is 65 cm^2 .

Calculate the value of r .

$r = \dots\dots\dots$ [4]

$$\begin{aligned} \mathbf{68} \quad \text{Segment} &= \frac{140}{360}\pi r^2 - \frac{1}{2}r^2 \sin 140^\circ = 1.2217r^2 - 0.3214r^2 = 0.9003r^2 \\ 0.9003r^2 &= 65 \Rightarrow r^2 = 72.19 \Rightarrow \boxed{r = 8.50} \end{aligned}$$

69

NOT TO
SCALE

OAB is the sector of a circle, centre O .
 $OB = 8$ cm and angle $AOB = 30^\circ$.
 BP is perpendicular to OA .

(a) Calculate AP .

$AP = \dots\dots\dots$ cm [3]

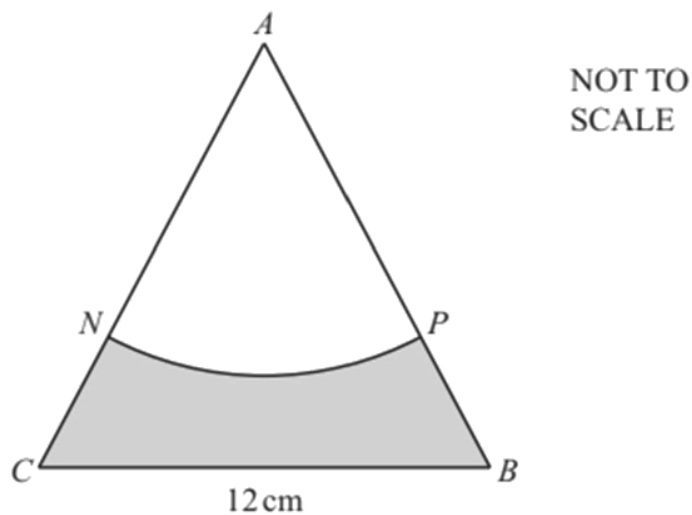
(b) Work out the area of the shaded region APB .

$\dots\dots\dots$ cm² [3]

69 (a) $OP = 8 \cos 30^\circ = 4\sqrt{3} = 6.928$; $AP = 8 - 6.928 = \boxed{1.07 \text{ cm}}$

(b) Sector $= \frac{30}{360}\pi(64) = 16.76$; triangle $OPB = \frac{1}{2} \times 6.928 \times 4 = 13.86$. Shaded = $\boxed{2.90 \text{ cm}^2}$

70



The diagram shows an equilateral triangle ABC with all sides of length 12 cm .

ANP is a sector of a circle, centre A .

N lies on AC such that $AN : NC = 2 : 1$.

P lies on AB such that $AP : PB = 2 : 1$.

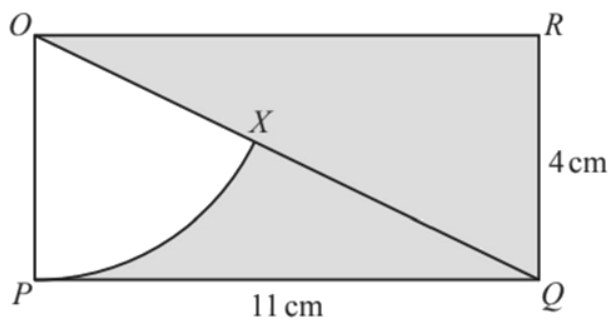
Calculate the area of the shaded region.

..... cm^2 [4]

70 $AN = AP = \frac{2}{3} \times 12 = 8$. Triangle $= \frac{1}{2}(12)(12) \sin 60^\circ = 62.35$; sector $= \frac{60}{360} \pi (8)^2 = 33.51$

Shaded $= 62.35 - 33.51 = \boxed{28.8 \text{ cm}^2}$

71.

NOT TO
SCALE

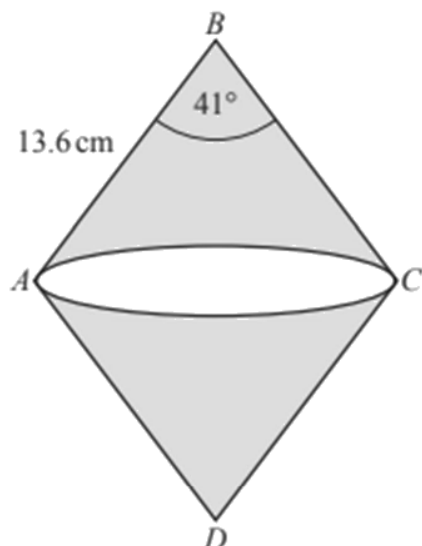
The diagram shows a rectangle $OPQR$ with length 11 cm and width 4 cm.
 OQ is a diagonal and OPX is a sector of a circle, centre O .

Calculate the percentage of the rectangle that is shaded.

..... % [5]

71 $\tan POQ = \frac{11}{4} \Rightarrow \angle POQ = 70.02^\circ$. Sector $OPX = \frac{70.02}{360} \pi (4)^2 = 9.78$
Shaded $= 44 - 9.78 = 34.22$; $\frac{34.22}{44} \times 100 = \boxed{77.8\%}$

72

NOT TO
SCALE

$ABCD$ is a rhombus with side length 13.6 cm .

Angle $ABC = 41^\circ$.

BAC is a sector of a circle with centre B .

DAC is a sector of a circle with centre D .

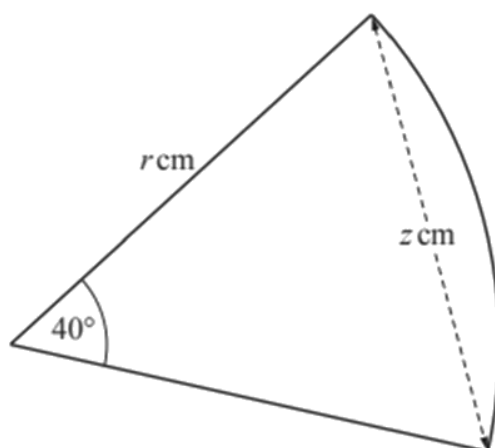
Calculate the shaded area.

..... cm^2 [4]

72 Triangle $ABC = \frac{1}{2}(13.6)^2 \sin 41^\circ = 60.67$; sector $= \frac{41}{360}\pi(13.6)^2 = 66.18$; segment $= 5.51$.

$$\text{Shaded} = 2 \times \text{triangle} - 2 \times \text{segment} = 121.34 - 11.01 = \boxed{110 \text{ cm}^2}$$

73 □

NOT TO
SCALE

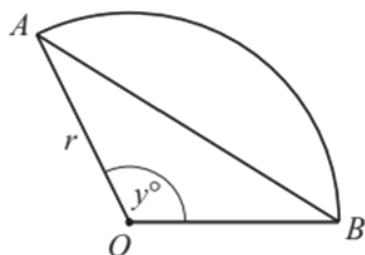
This sector of a circle has radius r and perimeter 20 cm.

Find the value of z .

$z = \dots\dots\dots$ [6]

$$\begin{aligned} \mathbf{73} \quad 2r + \frac{40}{360} \times 2\pi r &= 20 \Rightarrow r \left(2 + \frac{2\pi}{9} \right) = 20 \Rightarrow r = 7.4125 \\ z^2 &= r^2 + r^2 - 2r^2 \cos 40^\circ \Rightarrow \boxed{z = 5.07} \end{aligned}$$

74

NOT TO
SCALE

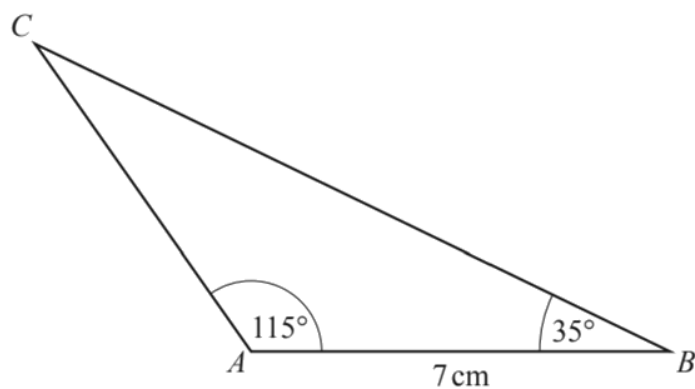
The diagram shows a sector, OAB , of a circle with centre O and radius r .
The area of triangle OAB is half the area of the sector.
Angle $AOB = y^\circ$ and is obtuse.

- (i) Show that $360 \sin y = \pi y$.

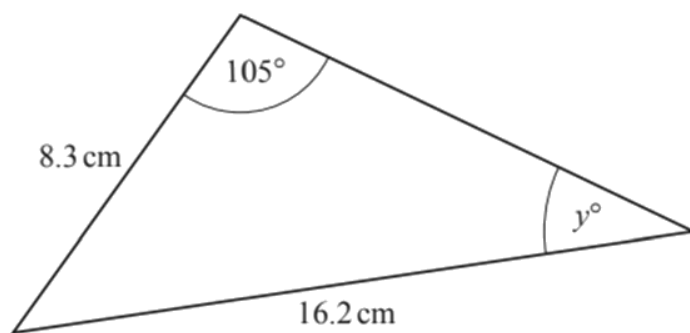
[2]

74 $\frac{1}{2}r^2 \sin y = \frac{1}{2} \times \frac{y}{360} \pi r^2 \Rightarrow \sin y = \frac{\pi y}{360} \Rightarrow 360 \sin y = \pi y \checkmark$

75

NOT TO
SCALECalculate the length BC . $BC = \dots\dots\dots$ cm [4]

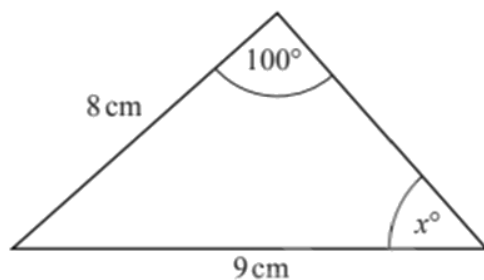
76

NOT TO
SCALECalculate the value of y . $y = \dots\dots\dots$ [3]

$$75 \quad \text{Angle } C = 180 - 115 - 35 = 30^\circ. \quad \frac{BC}{\sin 115^\circ} = \frac{7}{\sin 30^\circ} \Rightarrow BC = \boxed{12.7 \text{ cm}}$$

$$76 \quad \frac{\sin y}{8.3} = \frac{\sin 105^\circ}{16.2} \Rightarrow \sin y = 0.4949 \Rightarrow \boxed{y = 29.7}$$

77

NOT TO
SCALE

- (a) Calculate the value of x .

$x = \dots\dots\dots$ [3]

- (b) Calculate the area of the triangle.

$\dots\dots\dots \text{cm}^2$ [3]

- 78 In triangle PQR , $PQ = 9 \text{ cm}$, $QR = 13 \text{ cm}$ and angle $QPR = 72^\circ$.
Calculate angle PRQ .

Angle $PRQ = \dots\dots\dots$ [3]

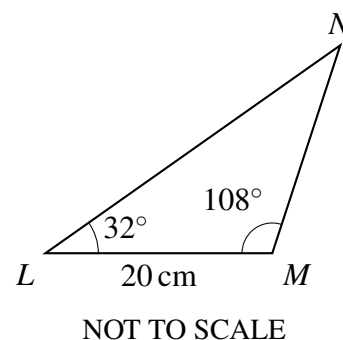
77 (a) $\frac{\sin x}{8} = \frac{\sin 100^\circ}{9} \Rightarrow \sin x = 0.8754 \Rightarrow \boxed{x = 61.1}$

(b) Third angle $= 180 - 100 - 61.1 = 18.9^\circ$ (between 8 and 9). Area $= \frac{1}{2}(8)(9) \sin 18.9^\circ = \boxed{11.7 \text{ cm}^2}$

78 $\frac{\sin R}{9} = \frac{\sin 72^\circ}{13} \Rightarrow \sin R = 0.6584 \Rightarrow \boxed{41.2^\circ}$ (unique, since $9 < 13$)

- 79** In triangle LMN , $LM = 20$ cm, angle $L = 32^\circ$ and angle $M = 108^\circ$.

- (a) Find angle N .
 (b) Calculate the length of the longest side.
 (c) Calculate the length of the shortest side.



- 80** In triangle XYZ , $XY = 7.4$ cm, angle $YXZ = 61^\circ$ and angle $XZY = 53^\circ$.
 Calculate (a) YZ , (b) XZ .

[5]

(a) cm (b) [4]

- 81** Prove the sine rule $\frac{a}{\sin A} = \frac{b}{\sin B}$ by drawing the perpendicular from C to AB and finding its length in two different ways.

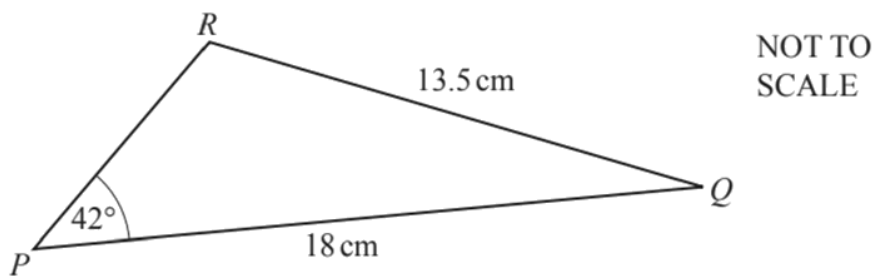
[3]

79 (a) $180 - 32 - 108 = \boxed{40^\circ}$ (b) opposite 108° : $LN = \frac{20 \sin 108^\circ}{\sin 40^\circ} = \boxed{29.6 \text{ cm}}$
(c) opposite 32° : $MN = \frac{20 \sin 32^\circ}{\sin 40^\circ} = \boxed{16.5 \text{ cm}}$

80 Angle $Y = 66^\circ$. (a) $YZ = \frac{7.4 \sin 61^\circ}{\sin 53^\circ} = \boxed{8.10 \text{ cm}}$ (b) $XZ = \frac{7.4 \sin 66^\circ}{\sin 53^\circ} = \boxed{8.46 \text{ cm}}$

81 Let the perpendicular be h . Triangle ACH : $h = b \sin A$; triangle BCH : $h = a \sin B$.
So $a \sin B = b \sin A \Rightarrow \frac{a}{\sin A} = \frac{b}{\sin B} \checkmark$

82



Calculate the obtuse angle PRQ .

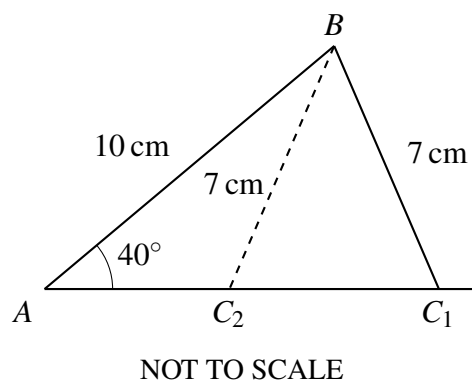
Angle $PRQ = \dots\dots\dots$ [4]

- 83** In triangle ABC , $AB = 10$ cm, $BC = 7$ cm and angle $BAC = 40^\circ$.

The diagram shows that two different triangles, ABC_1 and ABC_2 , are possible.

(a) Find the two possible values of angle ACB .

(b) Find the two possible lengths of AC .



[6]

$$\mathbf{82} \quad \frac{\sin R}{18} = \frac{\sin 42^\circ}{13.5} \Rightarrow \sin R = 0.8922 \Rightarrow R = 63.1^\circ \text{ (acute). Obtuse: } 180 - 63.1 = \boxed{116.9^\circ}$$

$$\mathbf{83} \quad (\text{a}) \sin C = \frac{10 \sin 40^\circ}{7} = 0.9183 \Rightarrow C = \boxed{66.7^\circ} \text{ or } \boxed{113.3^\circ} \text{ (both valid: } 40 + 113.3 < 180)$$

$$(\text{b}) \ C = 66.7^\circ: B = 73.3^\circ, AC = \frac{7 \sin 73.3^\circ}{\sin 40^\circ} = \boxed{10.4 \text{ cm}}; \ C = 113.3^\circ: B = 26.7^\circ, AC = \frac{7 \sin 26.7^\circ}{\sin 40^\circ} = \boxed{4.89 \text{ cm}}$$

- 84** In triangle PQR , $PQ = 16$ cm, $QR = 12$ cm and angle $QPR = 38^\circ$. Angle PRQ is obtuse. Calculate angle PRQ .

Angle $PRQ = \dots\dots\dots$ [4]

- 85** A student tries to draw a triangle with $BC = 4$ cm, $AC = 10$ cm and angle $BAC = 30^\circ$. Use the sine rule to show that no such triangle exists.

[2]

- 86** For each set of measurements, state how many different triangles ABC can be drawn. Show your working.
- (a) $A = 30^\circ$, $a = 5$ cm, $b = 10$ cm
 (b) $A = 50^\circ$, $a = 12$ cm, $b = 9$ cm
 (c) $A = 35^\circ$, $a = 6$ cm, $b = 9$ cm

(a) $\dots\dots\dots$ (b) $\dots\dots\dots$ (c) $\dots\dots\dots$ [4]

- 87** Explain why finding an angle with the cosine rule never gives an ambiguous answer, but finding an angle with the sine rule sometimes does.

[2]

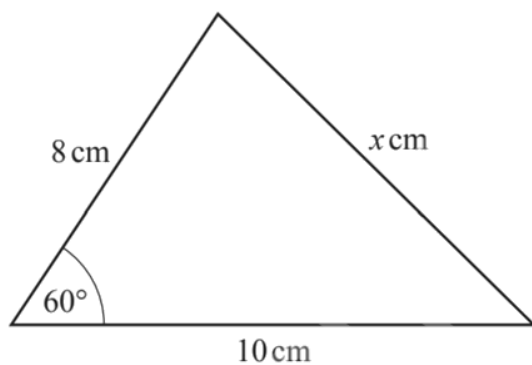
84 $\sin R = \frac{16 \sin 38^\circ}{12} = 0.8209 \Rightarrow 55.2^\circ$; obtuse: $\boxed{124.8^\circ}$

85 $\sin B = \frac{10 \sin 30^\circ}{4} = 1.25 > 1$, which is impossible, so **no triangle exists**.

86 (a) $\sin B = \frac{10 \sin 30^\circ}{5} = 1 \Rightarrow B = 90^\circ$: $\boxed{1}$ (b) $\sin B = 0.575 \Rightarrow 35.1^\circ$ or 144.9° ,
but $50 + 144.9 > 180$: $\boxed{1}$
(c) $\sin B = 0.860 \Rightarrow 59.4^\circ$ or 120.6° , $35 + 120.6 < 180$: $\boxed{2}$

87 Each value of $\cos A$ gives exactly one angle between 0° and 180° (negative $\Rightarrow$ obtuse). But $\sin \theta = \sin(180^\circ - \theta)$, so one value of $\sin A$ can give two angles.

88

NOT TO
SCALEFind the value of x^2 .

$$x^2 = \dots\dots\dots [3]$$

89 A triangle has sides of length 11 cm, 10 cm and 9 cm.

Calculate the largest angle in the triangle.

$$\dots\dots\dots [4]$$

88 $x^2 = 8^2 + 10^2 - 2(8)(10)\cos 60^\circ = 164 - 80 = \boxed{84}$

89 Largest angle opposite 11: $\cos \theta = \frac{10^2 + 9^2 - 11^2}{2(10)(9)} = \frac{60}{180} \Rightarrow \boxed{70.5^\circ}$

- 90** A triangle has sides of length 3 cm, 5 cm and 7 cm.

Work out the largest angle in the triangle.

..... [3]

- 91** A triangle has sides of length 8 cm and 11 cm with an included angle of 47° .
Calculate the length of the third side.

cm [3]

- 92** In triangle DEF , $DE = 4.5$ m, $DF = 6.8$ m and angle $EDF = 132^\circ$.
Calculate EF .

$EF =$ m [3]

90 Largest angle opposite 7: $\cos \theta = \frac{3^2 + 5^2 - 7^2}{2(3)(5)} = \frac{-15}{30} = -0.5 \Rightarrow \boxed{120^\circ}$

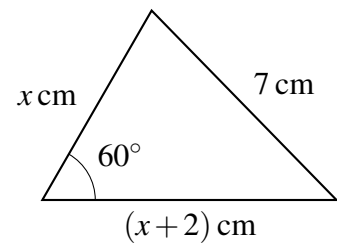
91 $x^2 = 8^2 + 11^2 - 2(8)(11)\cos 47^\circ = 64.97 \Rightarrow \boxed{8.06 \text{ cm}}$

92 $EF^2 = 20.25 + 46.24 - 61.2\cos 132^\circ = 107.44 \Rightarrow \boxed{10.4 \text{ m}}$ ($\cos 132^\circ < 0$, so add)

- 93** A triangle has sides 4 cm and 7 cm with an included angle of 120° .
Without using a calculator, find the exact length of the third side.

- 94** In the triangle shown, two sides are x cm and $(x + 2)$ cm, the angle between them is 60° and the third side is 7 cm.
- (a) Show that $x^2 + 2x - 45 = 0$.
- (b) Solve the equation to find x , correct to 2 decimal places.

cm [3]



NOT TO SCALE

- (b) $x =$ [5]
- 95** A parallelogram has sides of length 6 cm and 10 cm and an angle of 70° .
Calculate the lengths of both diagonals.

..... cm and [4]

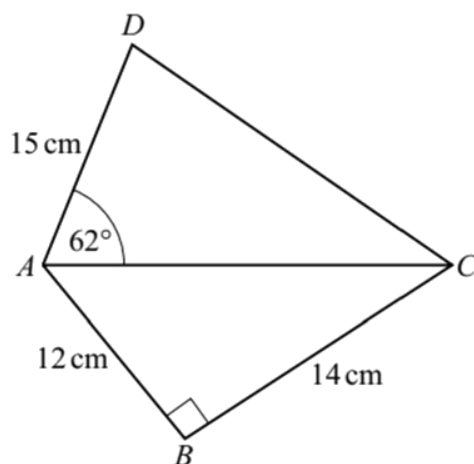
$$93 \quad x^2 = 16 + 49 - 56 \cos 120^\circ = 65 + 28 = 93 \Rightarrow \boxed{\sqrt{93} \text{ cm}}$$

$$94 \quad (a) \quad 49 = x^2 + (x+2)^2 - 2x(x+2) \cos 60^\circ = x^2 + 2x + 4 \Rightarrow x^2 + 2x - 45 = 0 \quad \checkmark$$

$$(b) \quad x = \frac{-2 + \sqrt{184}}{2} = \boxed{5.78}$$

$$95 \quad \text{Angles } 70^\circ \text{ and } 110^\circ. \quad d_1^2 = 136 - 120 \cos 70^\circ \Rightarrow \boxed{9.74 \text{ cm}}; \quad d_2^2 = 136 - 120 \cos 110^\circ \Rightarrow \boxed{13.3 \text{ cm}}$$

96

NOT TO
SCALE

The diagram shows a quadrilateral, $ABCD$, formed from two triangles, ABC and ACD . ABC is a right-angled triangle.

(a) Calculate angle BAC .

Angle $BAC = \dots\dots\dots$ [2]

(b) Calculate BD .

$BD = \dots\dots\dots$ cm [4]

(c) Calculate the shortest distance from D to AC .

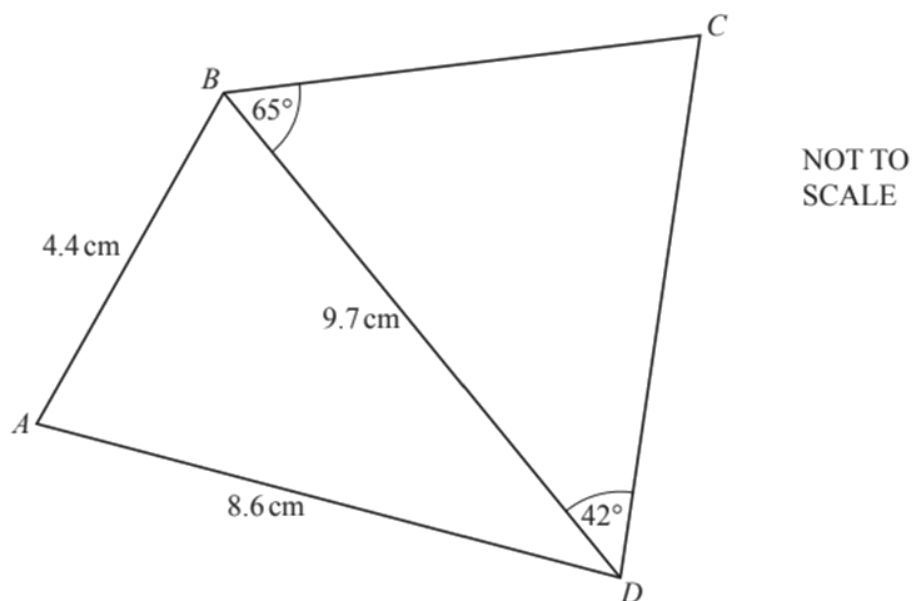
$\dots\dots\dots$ cm [3]

96 (a) $\cos BAC = \frac{12}{14} \Rightarrow \boxed{31.0^\circ}$

(b) $\angle BAD = 62 + 31.0 = 93.0^\circ$: $BD^2 = 12^2 + 15^2 - 2(12)(15)\cos 93.0^\circ \Rightarrow BD = \boxed{19.7 \text{ cm}}$

(c) $15 \sin 62^\circ = \boxed{13.2 \text{ cm}}$

97



- (a) Calculate angle ADB .

Angle $ADB = \dots\dots\dots$ [3]

- (b) Calculate DC .

$DC = \dots\dots\dots$ cm [4]

- (c) Calculate the shortest distance from C to BD .

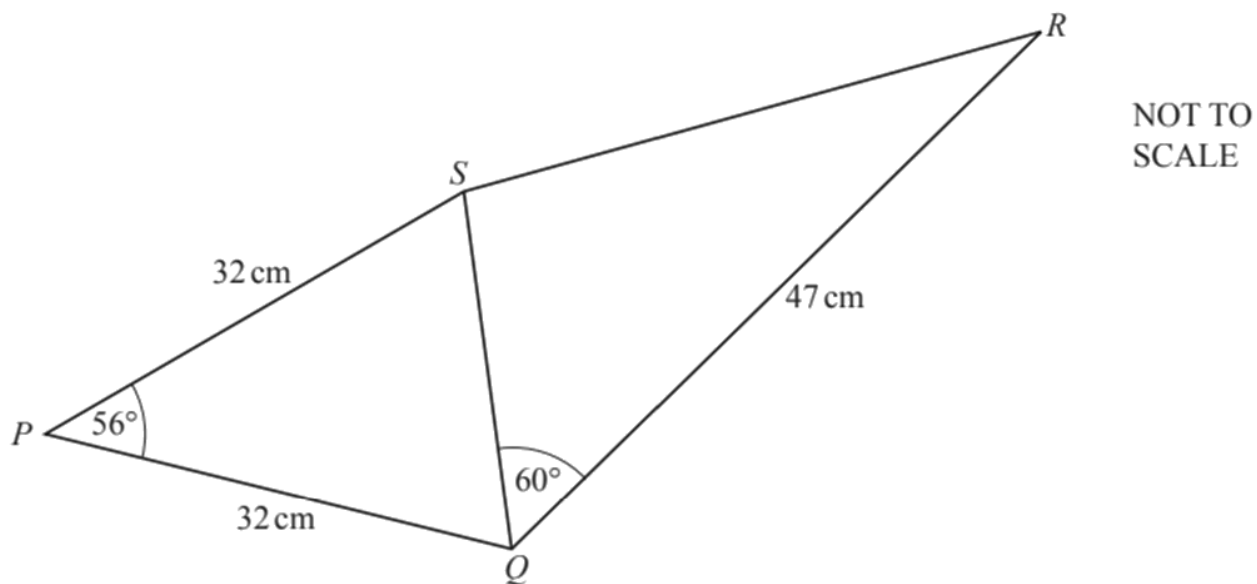
$\dots\dots\dots$ cm [3]

97 (a) $\cos ADB = \frac{8.6^2 + 9.7^2 - 4.4^2}{2(8.6)(9.7)} = 0.8913 \Rightarrow \boxed{27.0^\circ}$

(b) $\angle C = 180 - 65 - 42 = 73^\circ$: $DC = \frac{9.7 \sin 65^\circ}{\sin 73^\circ} = \boxed{9.19 \text{ cm}}$

(c) $9.19 \sin 42^\circ = \boxed{6.15 \text{ cm}}$

98



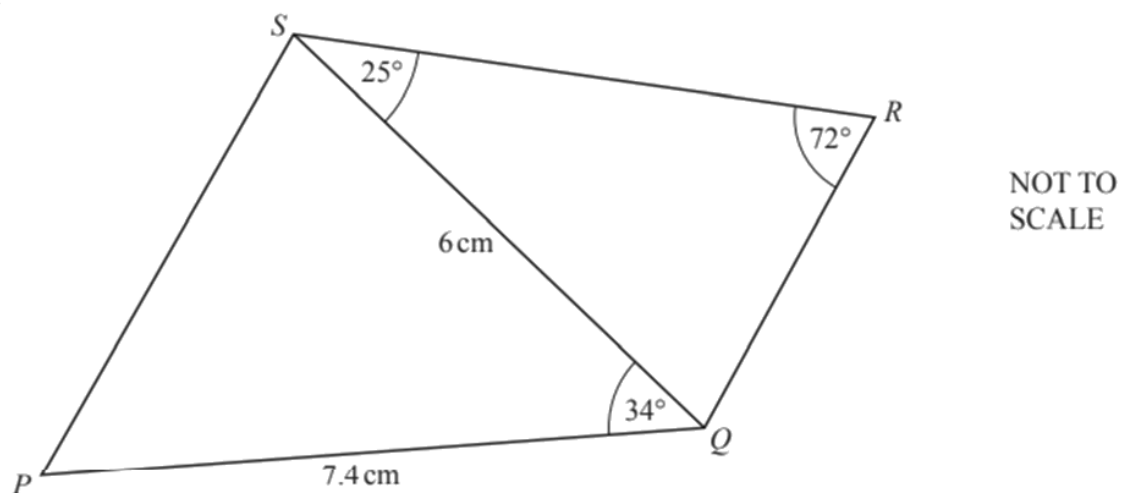
The diagram shows a quadrilateral $PQRS$ formed from two triangles, PQS and QRS . Triangle PQS is isosceles, with $PQ = PS = 32$ cm and angle $SPQ = 56^\circ$. $QR = 47$ cm and angle $SQR = 60^\circ$.

(i) Calculate SR .

$SR = \dots\dots\dots$ cm [4]

$$\begin{aligned} \mathbf{98} \quad \angle PQS = \angle PSQ = 62^\circ. \quad QS &= \frac{32 \sin 56^\circ}{\sin 62^\circ} = 30.05 \\ SR^2 &= 47^2 + 30.05^2 - 2(47)(30.05) \cos 60^\circ \Rightarrow SR = \boxed{41.2 \text{ cm}} \end{aligned}$$

99



The diagram shows a quadrilateral $PQRS$ formed from two triangles, PQS and QRS .

Calculate

(i) QR ,

$QR = \dots\dots\dots$ cm [3]

(ii) PS ,

$PS = \dots\dots\dots$ cm [3]

(iii) the area of quadrilateral $PQRS$.

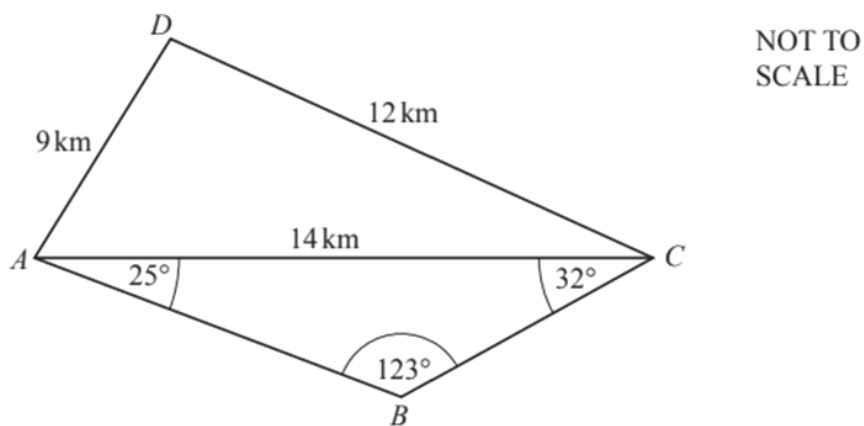
$\dots\dots\dots$ cm² [4]

$$99 \quad (i) \frac{QR}{\sin 25^\circ} = \frac{6}{\sin 72^\circ} \Rightarrow QR = \boxed{2.67 \text{ cm}}$$

$$(ii) PS^2 = 7.4^2 + 6^2 - 2(7.4)(6) \cos 34^\circ \Rightarrow PS = \boxed{4.14 \text{ cm}}$$

$$(iii) \angle SQR = 83^\circ. \text{ Area} = \frac{1}{2}(7.4)(6) \sin 34^\circ + \frac{1}{2}(6)(2.666) \sin 83^\circ = 12.41 + 7.94 = \boxed{20.4 \text{ cm}^2}$$

100



- (a) Calculate angle ACD .

Angle $ACD = \dots\dots\dots$ [4]

- (b) Show that $BC = 7.05$ km, correct to 2 decimal places.

[3]

- (c) Calculate the shortest distance from B to AC .

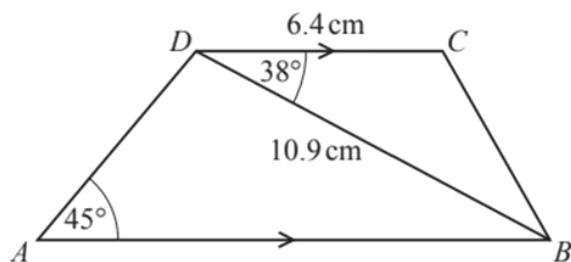
$\dots\dots\dots$ km [3]

$$100 \quad (a) \cos ACD = \frac{14^2 + 12^2 - 9^2}{2(14)(12)} = 0.7708 \Rightarrow \boxed{39.6^\circ}$$

$$(b) \frac{BC}{\sin 25^\circ} = \frac{14}{\sin 123^\circ} \Rightarrow BC = 7.0548 = 7.05 \text{ km } \checkmark$$

$$(c) 7.055 \sin 32^\circ = \boxed{3.74 \text{ km}}$$

101

NOT TO
SCALE

$ABCD$ is a trapezium with DC parallel to AB .

$DC = 6.4 \text{ cm}$, $DB = 10.9 \text{ cm}$, angle $CDB = 38^\circ$ and angle $DAB = 45^\circ$.

(a) Find CB .

$CB = \dots\dots\dots \text{ cm}$ [3]

(b) (i) Find angle ADB .

Angle $ADB = \dots\dots\dots$ [1]

(ii) Find AB .

$AB = \dots\dots\dots \text{ cm}$ [3]

(c) Calculate the area of the trapezium.

$\dots\dots\dots \text{ cm}^2$ [3]

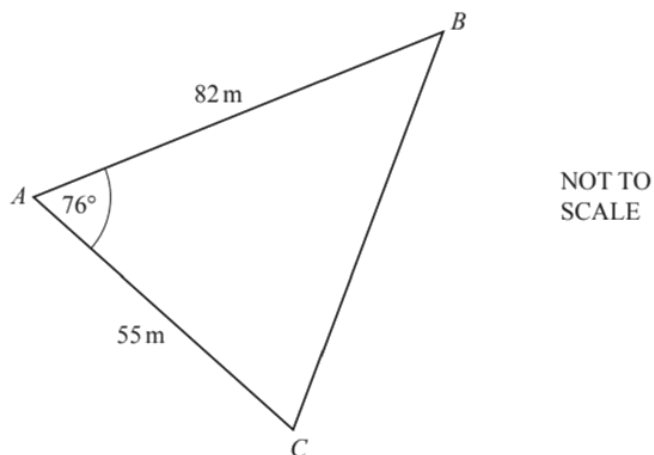
101 (a) $CB^2 = 6.4^2 + 10.9^2 - 2(6.4)(10.9)\cos 38^\circ \Rightarrow CB = \boxed{7.06 \text{ cm}}$

(b)(i) $\angle ABD = 38^\circ$ (alternate), so $\angle ADB = 180 - 45 - 38 = \boxed{97^\circ}$

(ii) $AB = \frac{10.9 \sin 97^\circ}{\sin 45^\circ} = \boxed{15.3 \text{ cm}}$

(c) Height $= 10.9 \sin 38^\circ = 6.711$; area $= \frac{1}{2}(6.4 + 15.30) \times 6.711 = \boxed{72.8 \text{ cm}^2}$

102



The diagram shows a field ABC .

(a) Calculate BC .

$BC = \dots\dots\dots\text{ m}$ [3]

(b) Calculate angle ACB .

Angle $ACB = \dots\dots\dots$ [3]

(c) A gate, G , lies on AB at the shortest distance from C .

Calculate AG .

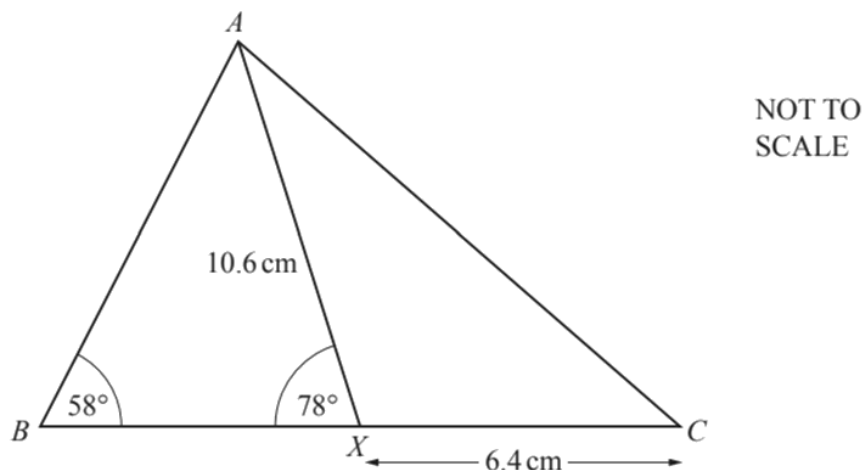
$AG = \dots\dots\dots\text{ m}$ [3]

$$\mathbf{102} \quad (\text{a}) \ BC^2 = 82^2 + 55^2 - 2(82)(55)\cos 76^\circ \Rightarrow BC = \boxed{87.0 \text{ m}}$$

$$(\text{b}) \ \sin C = \frac{82 \sin 76^\circ}{86.99} \Rightarrow \boxed{66.2^\circ}$$

$$(\text{c}) \ AG = 55 \cos 76^\circ = \boxed{13.3 \text{ m}}$$

103



The diagram shows triangle ABC .

X is a point on BC .

$AX = 10.6\text{ cm}$, $XC = 6.4\text{ cm}$, angle $ABC = 58^\circ$ and angle $AXB = 78^\circ$.

(a) Calculate AC .

$AC = \dots\dots\dots\text{ cm}$ [4]

(b) Calculate BX .

$BX = \dots\dots\dots\text{ cm}$ [4]

(c) Calculate the area of triangle ABC .

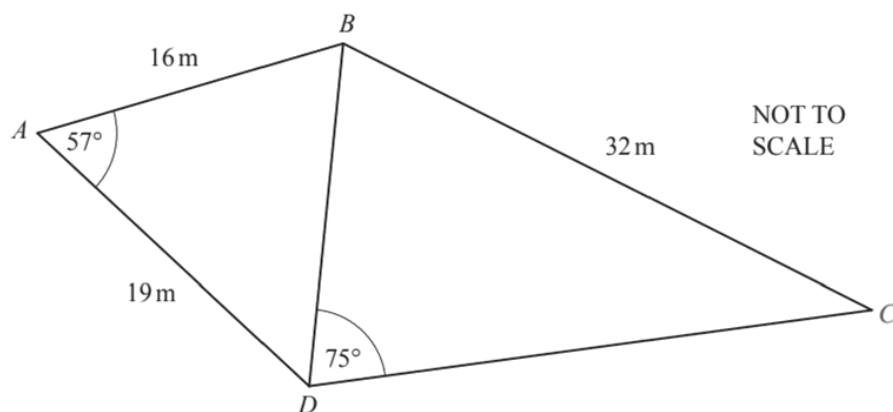
$\dots\dots\dots\text{ cm}^2$ [3]

103 (a) $\angle AXC = 102^\circ$: $AC^2 = 10.6^2 + 6.4^2 - 2(10.6)(6.4)\cos 102^\circ \Rightarrow AC = \boxed{13.5 \text{ cm}}$

(b) $\angle BAX = 180 - 58 - 78 = 44^\circ$: $BX = \frac{10.6 \sin 44^\circ}{\sin 58^\circ} = \boxed{8.68 \text{ cm}}$

(c) $\frac{1}{2}(8.683 + 6.4)(10.6) \sin 78^\circ = \boxed{78.2 \text{ cm}^2}$

104



The diagram shows a quadrilateral $ABCD$ made from two triangles, ABD and BCD .

- (a) Show that $BD = 16.9\text{ m}$, correct to 1 decimal place.

[3]

- (b) Calculate angle CBD .

Angle $CBD = \dots\dots\dots$ [4]

- (c) Find the area of the quadrilateral $ABCD$.

$\dots\dots\dots \text{ m}^2$ [3]

- (d) Find the shortest distance from B to AD .

$\dots\dots\dots \text{ m}$ [3]

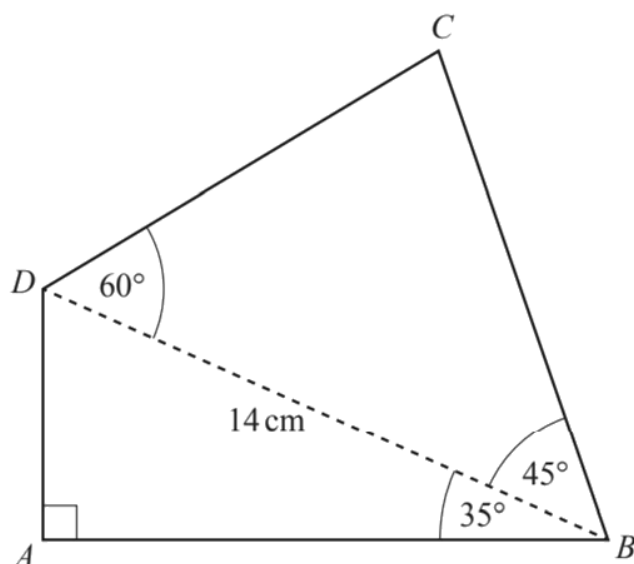
104 (a) $BD^2 = 16^2 + 19^2 - 2(16)(19)\cos 57^\circ = 285.86 \Rightarrow BD = 16.9 \text{ m} \checkmark$

(b) $\sin C = \frac{16.91 \sin 75^\circ}{32} \Rightarrow C = 30.7^\circ; \angle CBD = 180 - 75 - 30.7 = \boxed{74.3^\circ}$

(c) $\frac{1}{2}(16)(19)\sin 57^\circ + \frac{1}{2}(16.91)(32)\sin 74.3^\circ = 127.5 + 260.4 = \boxed{388 \text{ m}^2}$

(d) $16 \sin 57^\circ = \boxed{13.4 \text{ m}}$

105

NOT TO
SCALE

Calculate the perimeter of the quadrilateral $ABCD$.

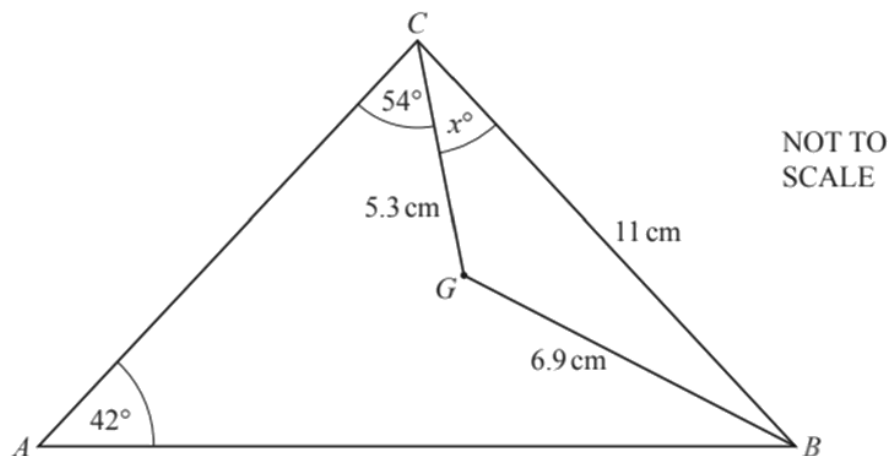
..... cm [7]

105 $AD = 14 \sin 35^\circ = 8.030$, $AB = 14 \cos 35^\circ = 11.468$. $\angle C = 180 - 60 - 45 = 75^\circ$

$$DC = \frac{14 \sin 45^\circ}{\sin 75^\circ} = 10.249, \quad BC = \frac{14 \sin 60^\circ}{\sin 75^\circ} = 12.552$$

$$\text{Perimeter} = 8.030 + 11.468 + 12.552 + 10.249 = \boxed{42.3 \text{ cm}}$$

106



The diagram shows triangle ABC with point G inside.
 $CB = 11$ cm, $CG = 5.3$ cm and $BG = 6.9$ cm.
Angle $CAB = 42^\circ$ and angle $ACG = 54^\circ$.

(i) Calculate the value of x .

$x = \dots\dots\dots$ [4]

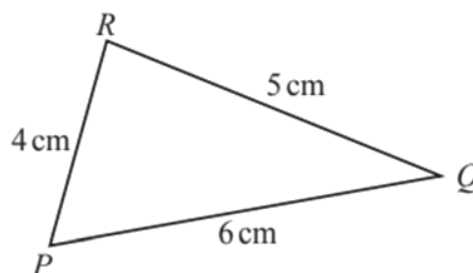
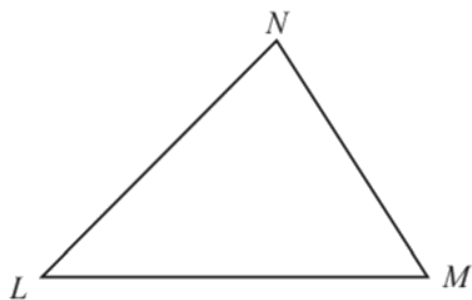
(ii) Calculate AC .

$AC = \dots\dots\dots$ cm [4]

$$106 \quad (i) \cos x = \frac{5.3^2 + 11^2 - 6.9^2}{2(5.3)(11)} = 0.8704 \Rightarrow \boxed{x = 29.5}$$

$$(ii) \angle ACB = 54 + 29.5 = 83.5^\circ, \angle B = 180 - 42 - 83.5 = 54.5^\circ. \quad AC = \frac{11 \sin 54.5^\circ}{\sin 42^\circ} = \boxed{13.4 \text{ cm}}$$

107

NOT TO
SCALE

In triangle LMN , the ratio angle L : angle M : angle $N = 4 : 5 : 6$.

In triangle PQR , $PQ = 6$ cm, $PR = 4$ cm and $QR = 5$ cm.

Calculate the difference between the largest angle in triangle PQR and the largest angle in triangle LMN .

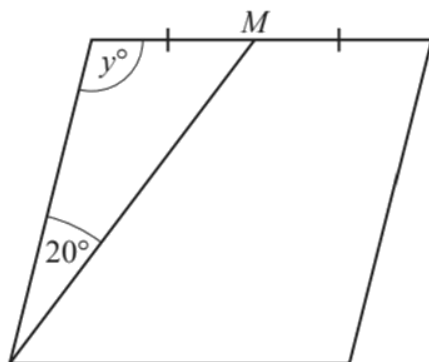
..... [7]

107 LMN : $4 + 5 + 6 = 15$ parts, $180 \div 15 = 12$, so angles $48^\circ, 60^\circ, 72^\circ$ — largest 72° .

PQR : largest angle opposite 6: $\cos R = \frac{4^2 + 5^2 - 6^2}{2(4)(5)} = \frac{1}{8} \Rightarrow 82.8^\circ$. Difference =

10.8°

108

NOT TO
SCALE

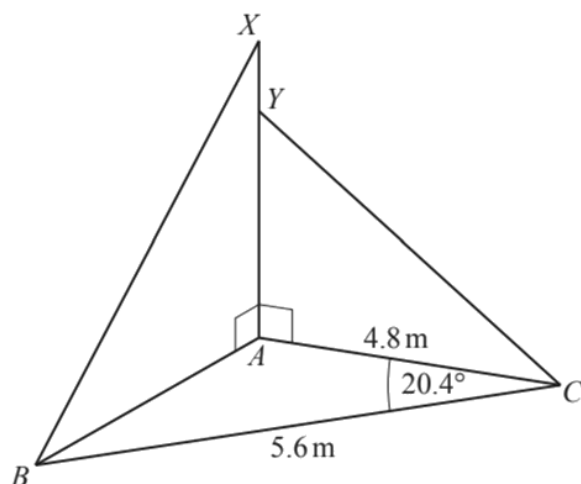
This rhombus has perimeter 20 cm and angle y is obtuse.
 M is the midpoint of one of the sides.

Find the value of y .

$y = \dots\dots\dots$ [5]

108 Side = $20 \div 4 = 5$; half side = 2.5. In the small triangle: $\frac{\sin \theta}{5} = \frac{\sin 20^\circ}{2.5} \Rightarrow$
 $\theta = 43.2^\circ$ (opposite 5)
 $y = 180 - 20 - 43.2 = \boxed{116.8^\circ}$

109

NOT TO
SCALE

ABC is a scalene triangle on horizontal ground.
 AYX is a straight vertical post, held in place by two straight wires XB and YC .
 $AC = 4.8$ m, $BC = 5.6$ m and angle $ACB = 20.4^\circ$.

(i) Calculate AB .

$AB = \dots\dots\dots$ m [3]

(ii) Angle $XBA = 64^\circ$.

Calculate AX .

$AX = \dots\dots\dots$ m [2]

(iii) $AY = 2.9$ m.

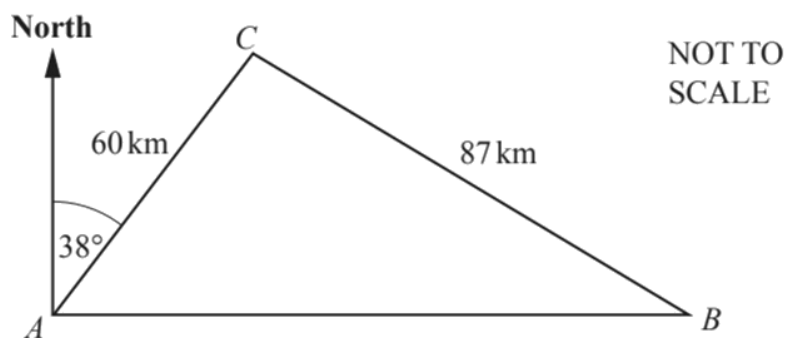
Calculate the area of triangle YAC .

$\dots\dots\dots$ m² [2]

109 (i) $AB^2 = 4.8^2 + 5.6^2 - 2(4.8)(5.6)\cos 20.4^\circ \Rightarrow AB = \boxed{2.00 \text{ m}}$

(ii) $AX = AB \tan 64^\circ = \boxed{4.11 \text{ m}}$ (iii) Right angle at A: $\frac{1}{2} \times 4.8 \times 2.9 = \boxed{6.96 \text{ m}^2}$

110



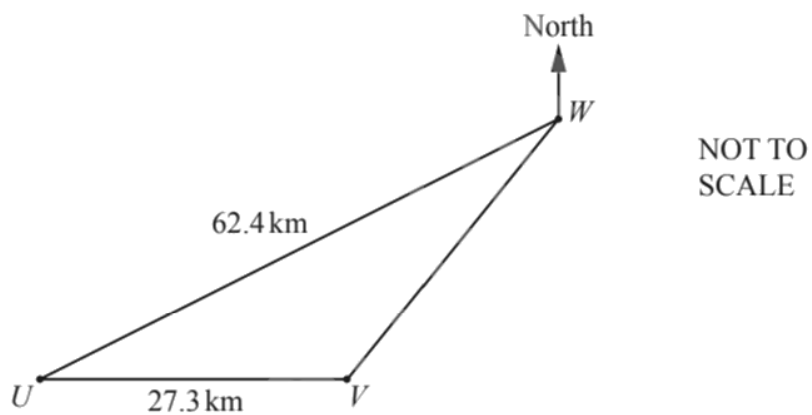
The diagram shows the straight roads between town A , town B and town C .
 $AC = 60$ km, $CB = 87$ km and B is due east of A .
The bearing of C from A is 038° .

- (a) Show that angle $ACB = 95.1^\circ$, correct to 1 decimal place.

[5]

110 $\angle CAB = 90 - 38 = 52^\circ$. $\sin B = \frac{60 \sin 52^\circ}{87} \Rightarrow \angle ABC = 32.9^\circ$
 $\angle ACB = 180 - 52 - 32.9 = 95.1^\circ \checkmark$

111



The diagram shows the position of three towns, U , V and W .
 U is due west of V and angle $UVW = 125^\circ$.

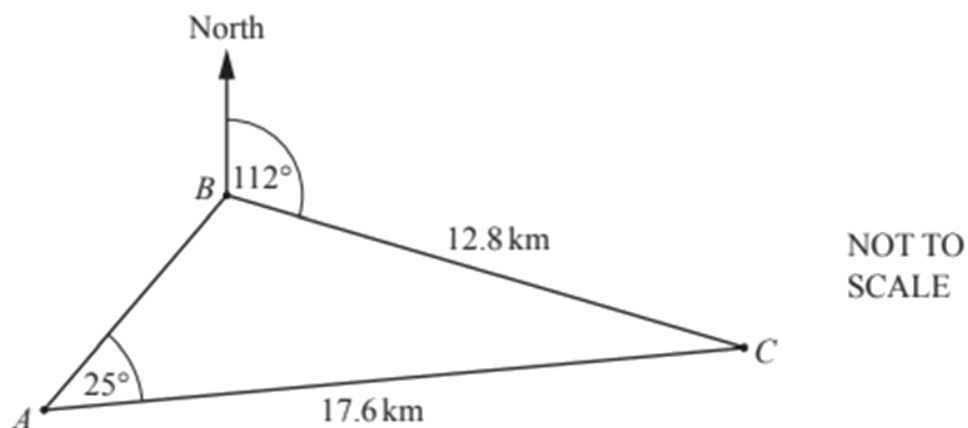
Calculate the bearing of U from W .

..... [4]

111 Sine rule: $\sin W = \frac{27.3 \sin 125^\circ}{62.4} \Rightarrow \angle UWV = 21.0^\circ$, so $\angle WUV = 180 - 125 - 21.0 = 34.0^\circ$.

Bearing of W from $U = 90 - 34.0 = 056^\circ$, so bearing of U from $W = 056 + 180 = \boxed{236^\circ}$

112



The diagram shows the positions of three ships A , B and C .
 $AC = 17.6$ km, $BC = 12.8$ km and angle $BAC = 25^\circ$.
The bearing of C from B is 112° and angle ABC is obtuse.

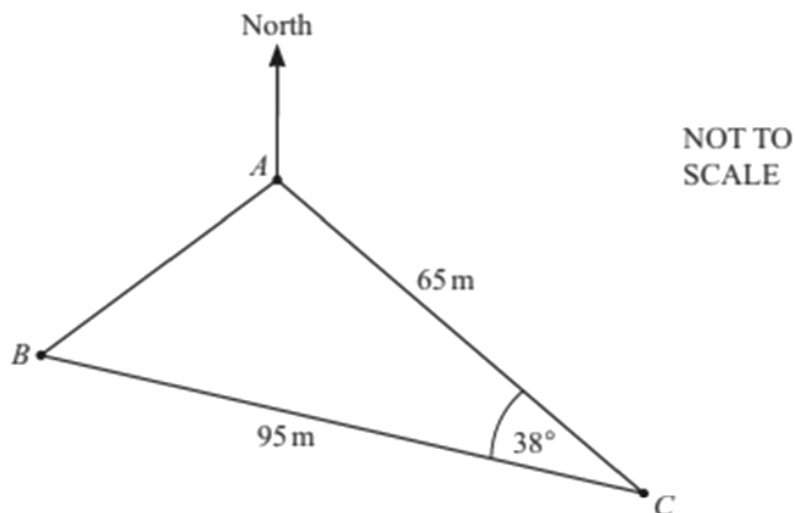
Calculate the bearing of B from A .

..... [5]

112 $\sin B = \frac{17.6 \sin 25^\circ}{12.8} = 0.5811 \Rightarrow 35.5^\circ$; obtuse so $\angle ABC = 144.5^\circ$.

Bearing of A from B = $112 + 144.5 = 256.5^\circ$, so bearing of B from A = $256.5 - 180 = 076.5^\circ$

113



The diagram shows three points A , B and C on horizontal ground.
 The bearing of C from A is 145° and angle $ACB = 38^\circ$.
 $AC = 65$ m and $BC = 95$ m.

- (a) Find the bearing of B from C .

..... [2]

- (b) Show that $AB = 59.3$ m, correct to 1 decimal place.

[3]

- (c) Angle BAC is obtuse.

Work out the bearing of B from A .

..... [4]

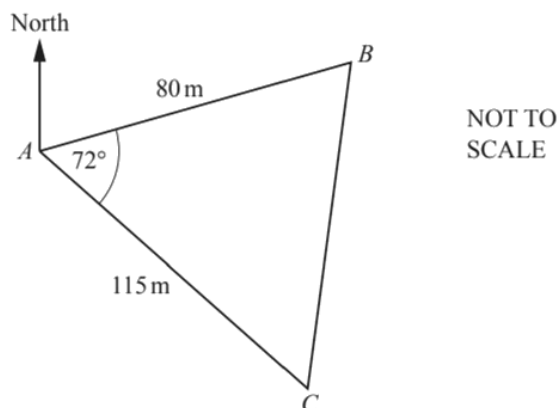
113 (a) Bearing of A from $C = 325^\circ$; B is 38° further anticlockwise: $325 - 38 = \boxed{287^\circ}$

(b) $AB^2 = 65^2 + 95^2 - 2(65)(95)\cos 38^\circ \Rightarrow AB = 59.31 = 59.3 \text{ m } \checkmark$

(c) $\sin A = \frac{95 \sin 38^\circ}{59.31} = 0.9861 \Rightarrow 80.4^\circ$; angle BAC is obtuse $\Rightarrow 99.6^\circ$.

Bearing of B from $A = 145 + 99.6 = \boxed{244.6^\circ}$

114



The diagram shows the positions of three points A , B and C in a field.

- (a) Show that BC is 118.1 m, correct to 1 decimal place.

[3]

- (b) Calculate angle ABC .

Angle $ABC = \dots\dots\dots$ [3]

- (c) The bearing of C from A is 147° .

Find the bearing of

- (i) A from B ,

$\dots\dots\dots$ [3]

- (ii) B from C .

$\dots\dots\dots$ [2]

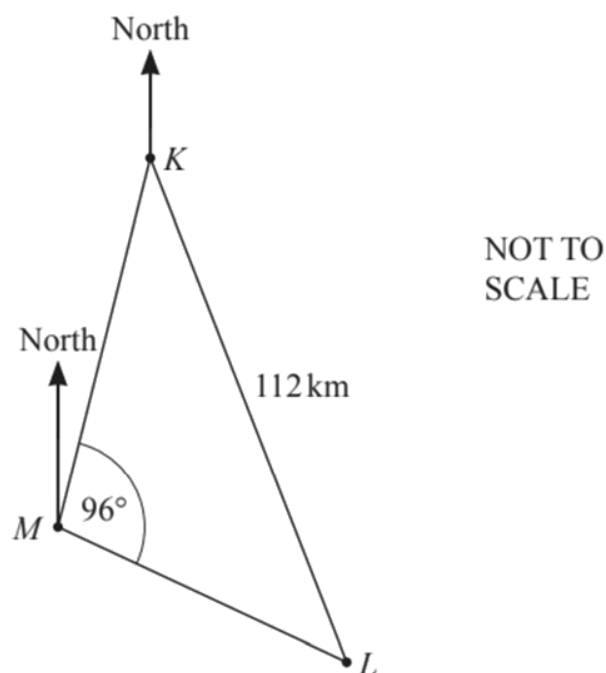
114 (a) $BC^2 = 80^2 + 115^2 - 2(80)(115)\cos 72^\circ \Rightarrow BC = 118.06 = 118.1 \text{ m } \checkmark$

(b) $\sin B = \frac{115 \sin 72^\circ}{118.06} \Rightarrow \boxed{67.9^\circ}$

(c)(i) Bearing of B from $A = 147 - 72 = 075^\circ$, so A from $B = \boxed{255^\circ}$

(ii) $\angle ACB = 180 - 72 - 67.9 = 40.1^\circ$; A from $C = 327^\circ$; B from $C = 327 + 40.1 - 360 = \boxed{007^\circ}$

115



The diagram shows the positions of a lighthouse, L , and two ships, K and M .
 The bearing of L from K is 155° and $KL = 112$ km.
 The bearing of K from M is 010° and angle $KML = 96^\circ$.

Find the bearing and distance of ship M from the lighthouse, L .

Bearing

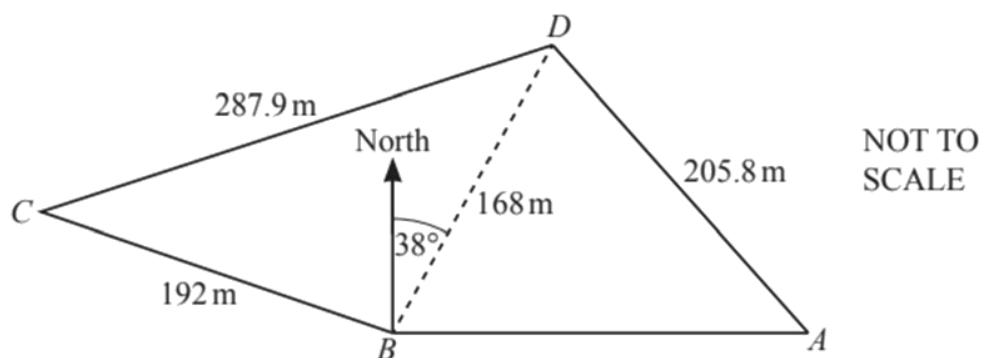
Distance km [5]

115 At K : bearing to $M = 190^\circ$, to $L = 155^\circ$, so $\angle MKL = 35^\circ$; $\angle KLM = 180 - 96 - 35 = 49^\circ$.

Bearing of K from $L = 335^\circ$, so M from $L = 335 - 49 = \boxed{286^\circ}$

$$ML = \frac{112 \sin 35^\circ}{\sin 96^\circ} = \boxed{64.6 \text{ km}}$$

116



The diagram shows a field, $ABCD$, on horizontal ground.
 $BC = 192$ m, $CD = 287.9$ m, $BD = 168$ m and $AD = 205.8$ m.

- (a) (i) Calculate angle CBD and show that it rounds to 106.0° , correct to 1 decimal place.

[4]

- (ii) The bearing of D from B is 038° .

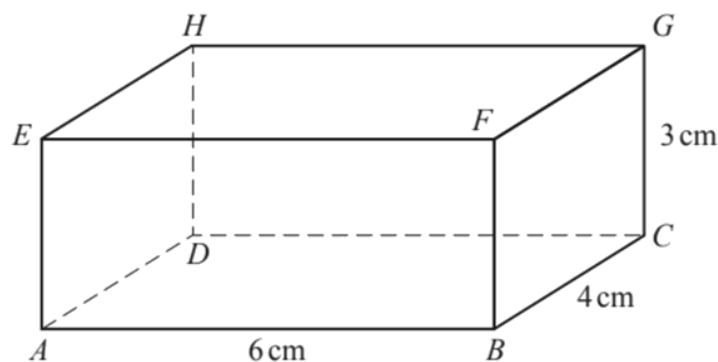
Find the bearing of C from B .

..... [1]

$$\mathbf{116} \quad (\text{i}) \cos CBD = \frac{192^2 + 168^2 - 287.9^2}{2(192)(168)} = -0.2759 \Rightarrow 106.0^\circ \checkmark$$

$$(\text{ii}) C \text{ is } 106^\circ \text{ anticlockwise from } D: 38 - 106 = -68 \Rightarrow \boxed{292^\circ}$$

117


NOT TO
SCALE

$ABCDEFGH$ is a cuboid.

Find the length of AG .

Give your answer in surd form.

..... cm [3]

118 A cuboid measures 24 cm by 12 cm by 8 cm.

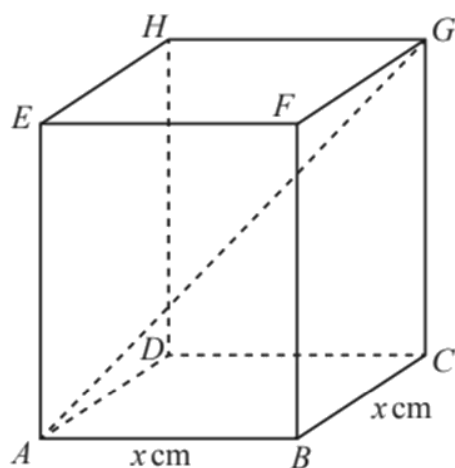
Calculate the length of a diagonal of the cuboid.

..... cm [3]

$$117 \quad AG = \sqrt{6^2 + 4^2 + 3^2} = \boxed{\sqrt{61} \text{ cm}}$$

$$118 \quad \sqrt{24^2 + 12^2 + 8^2} = \sqrt{784} = \boxed{28 \text{ cm}}$$

119

NOT TO
SCALE

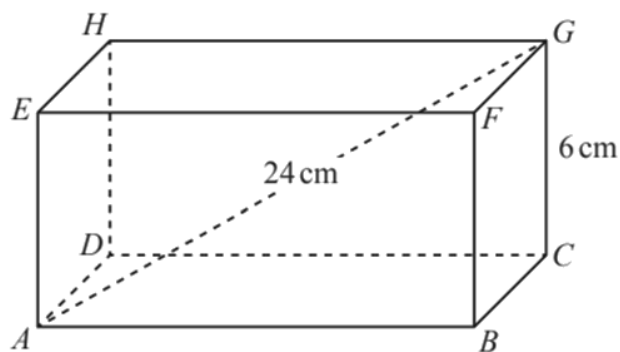
$ABCDEFGH$ is a cuboid with a square base of side x cm.
 $CG = 20$ cm and $AG = 28$ cm.

Calculate the value of x .

$x = \dots\dots\dots$ [4]

119 $x^2 + x^2 + 20^2 = 28^2 \Rightarrow 2x^2 = 384 \Rightarrow x^2 = 192 \Rightarrow \boxed{x = 13.9} \quad (8\sqrt{3})$

120

NOT TO
SCALE

The diagram shows a cuboid $ABCDEFGH$.
 $CG = 6$ cm, $AG = 24$ cm and $AB = 2BC$.

(a) Calculate AB .

$AB = \dots\dots\dots$ cm [4]

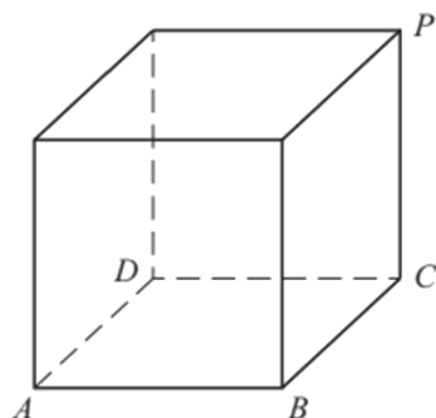
(b) Calculate the angle between AG and the base $ABCD$.

$\dots\dots\dots$ [3]

120 (a) $AC^2 = 24^2 - 6^2 = 540$; $(2x)^2 + x^2 = 540 \Rightarrow x^2 = 108 \Rightarrow BC = 10.39$, $AB =$
 $\boxed{20.8 \text{ cm}}$

(b) $\sin \theta = \frac{6}{24} \Rightarrow \boxed{14.5^\circ}$

121

NOT TO
SCALE

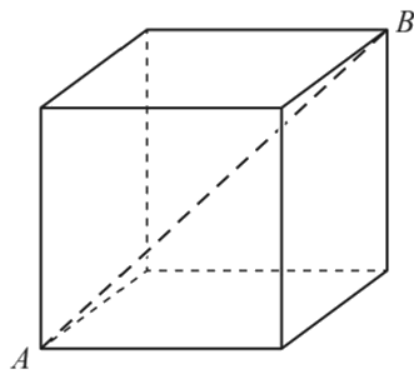
The diagram shows a cube.

Calculate the angle between the diagonal AP and the base $ABCD$.

..... [4]

121 Let edge = 1: $AC = \sqrt{2}$, $CP = 1$. $\tan \theta = \frac{1}{\sqrt{2}} \Rightarrow \boxed{35.3^\circ}$

122

NOT TO
SCALE

The diagram shows a cube.
The length of the diagonal AB is 8.5 cm.

- (i) Calculate the length of an edge of the cube.

..... cm [3]

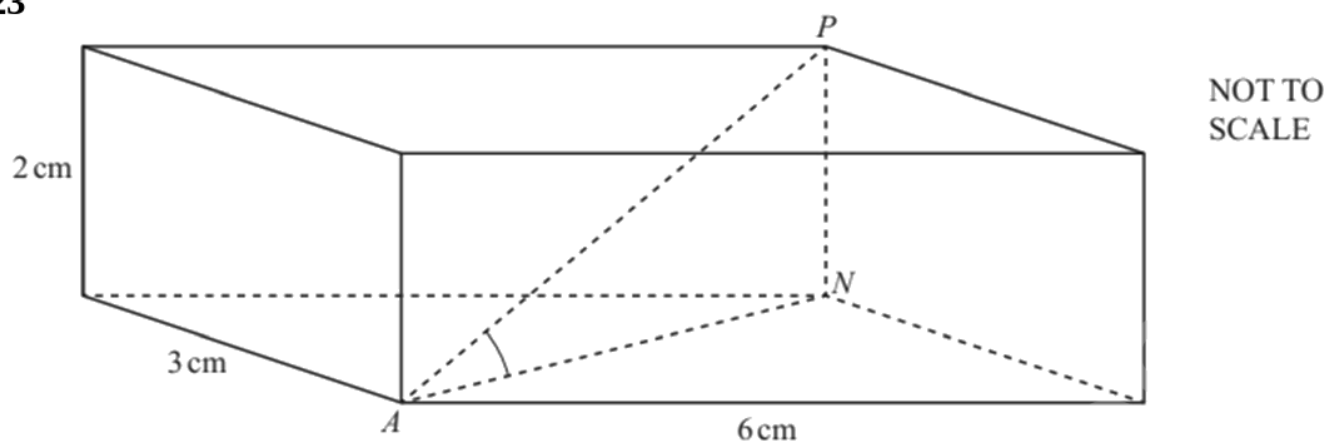
- (ii) Calculate the angle between AB and the base of the cube.

..... [3]

$$122 \quad (i) \quad 3x^2 = 8.5^2 \Rightarrow x = \frac{8.5}{\sqrt{3}} = \boxed{4.91 \text{ cm}}$$

$$(ii) \quad \sin \theta = \frac{4.907}{8.5} \Rightarrow \boxed{35.3^\circ}$$

123



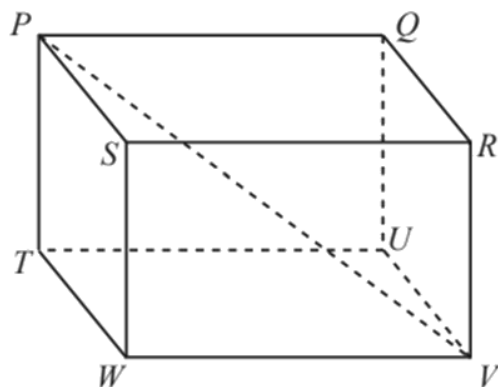
The diagram shows a cuboid measuring 6 cm by 3 cm by 2 cm.

Find the sine of the angle PAN , the angle between the diagonal PA and the base of the cuboid.

Sine of angle PAN = [4]

123 $AN = \sqrt{6^2 + 3^2} = \sqrt{45}$; $PA = \sqrt{45 + 2^2} = 7$. $\sin PAN = \frac{PN}{PA} = \frac{2}{7} \Rightarrow \angle PAN = \sin^{-1}\left(\frac{2}{7}\right) = 16.6^\circ$

124

NOT TO
SCALE

The diagram shows a cuboid $PQRSTUWV$.

$PV = 17.2$ cm

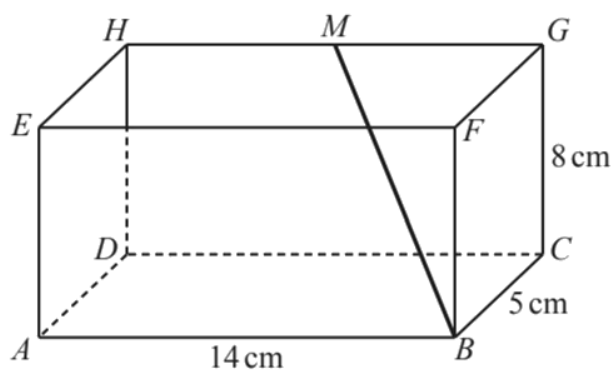
The angle between the line PV and the base $TUVW$ of the cuboid is 43° .

Calculate PT .

$PT = \dots\dots\dots$ cm [3]

124 Triangle PTV is right-angled at T : $PT = 17.2 \sin 43^\circ = \boxed{11.7 \text{ cm}}$

125

NOT TO
SCALE

The diagram shows a cuboid $ABCDEFGH$.

$AB = 14$ cm, $BC = 5$ cm and $CG = 8$ cm.

M is the midpoint of HG .

(a) Calculate BM .

..... cm [3]

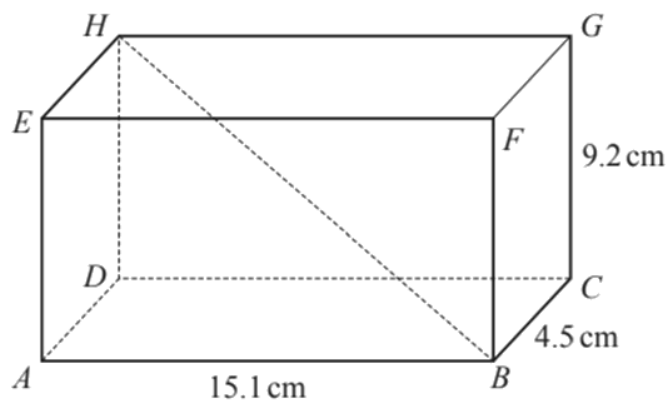
(b) Calculate the angle that BM makes with the base $ABCD$.

..... [3]

125 (a) Drop M to N on DC : $NC = 7$; $NB = \sqrt{7^2 + 5^2}$; $BM = \sqrt{5^2 + 7^2 + 8^2} = \sqrt{138} = \boxed{11.7 \text{ cm}}$

(b) $\sin \theta = \frac{8}{\sqrt{138}} \Rightarrow \boxed{42.9^\circ}$

126

NOT TO
SCALE

The diagram shows a cuboid $ABCDEFGH$.
 $AB = 15.1\text{ cm}$, $BC = 4.5\text{ cm}$ and $CG = 9.2\text{ cm}$.

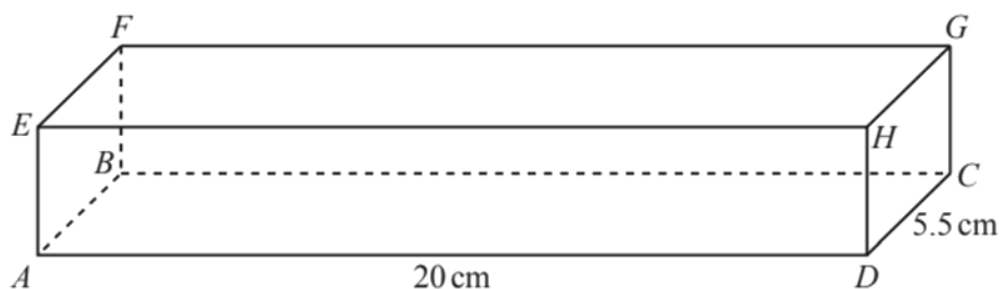
Calculate the angle that the diagonal BH makes with the face $ADHE$.

..... [4]

126 BA is perpendicular to face $ADHE$, so the angle is $\angle BHA$.

$$AH = \sqrt{4.5^2 + 9.2^2} = 10.24; \tan \theta = \frac{15.1}{10.24} \Rightarrow \boxed{55.9^\circ}$$

127

NOT TO
SCALE

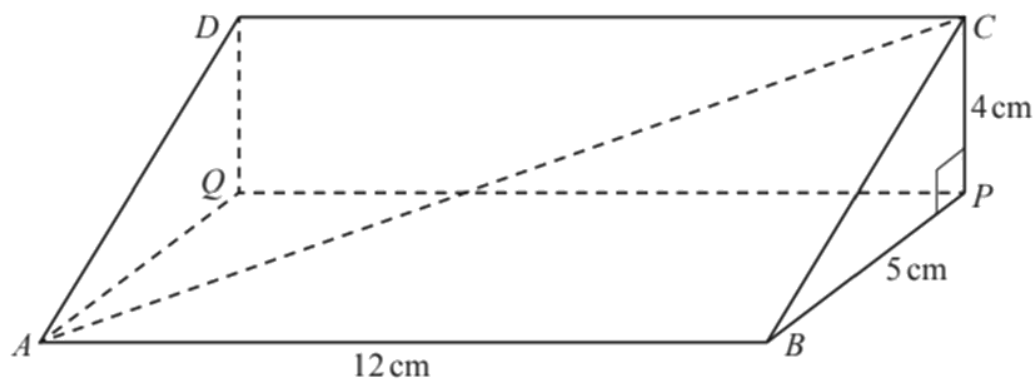
The diagram shows cuboid $ABCDEFGH$ of length 20 cm and width 5.5 cm.
The volume of the cuboid is 495 cm^3 .

Find the angle between the line AG and the base of the cuboid $ABCD$.

..... [5]

$$\mathbf{127} \quad \text{Height} = \frac{495}{20 \times 5.5} = 4.5. \quad AC = \sqrt{20^2 + 5.5^2} = 20.74$$
$$\tan \theta = \frac{4.5}{20.74} \Rightarrow \boxed{12.2^\circ}$$

128

NOT TO
SCALE

The diagram shows a triangular prism.
Angle $BPC = 90^\circ$.

(a) Calculate AC .

$AC = \dots\dots\dots$ cm [3]

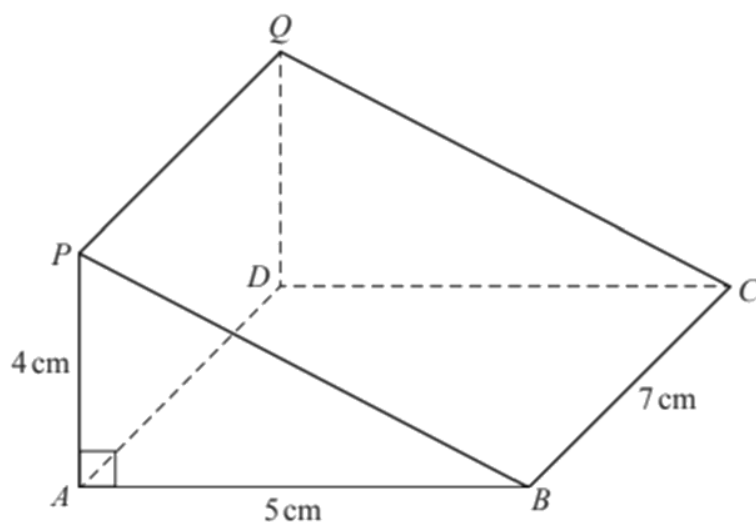
(b) Calculate the angle between AC and the base $ABPQ$.

$\dots\dots\dots$ [3]

128 (a) $BC = \sqrt{5^2 + 4^2} = \sqrt{41}$; $AC = \sqrt{12^2 + 41} = \sqrt{185} = \boxed{13.6 \text{ cm}}$

(b) $AP = \sqrt{12^2 + 5^2} = 13$, $CP = 4$: $\tan \theta = \frac{4}{13} \Rightarrow \boxed{17.1^\circ}$

129

NOT TO
SCALE

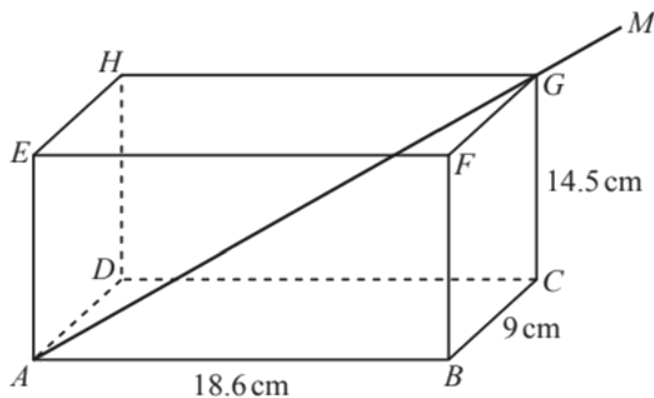
The diagram shows a triangular prism $ABCDQP$ of length 7 cm.
The cross-section is triangle PAB with $PA = 4$ cm, $AB = 5$ cm and angle $PAB = 90^\circ$.

Calculate the angle between the line PC and the base $ABCD$.

..... [4]

$$129 \quad AC = \sqrt{5^2 + 7^2} = \sqrt{74}. \quad \tan \theta = \frac{PA}{AC} = \frac{4}{\sqrt{74}} \Rightarrow \boxed{24.9^\circ}$$

130

NOT TO
SCALE

The diagram shows an open rectangular box $ABCDEFGH$.

$AB = 18.6$ cm, $BC = 9$ cm and $CG = 14.5$ cm.

A straight stick AGM rests against A and G and extends outside the box to M .

- (a) Calculate the angle between the stick and the base of the box.

..... [4]

- (b) $AM = 30$ cm.

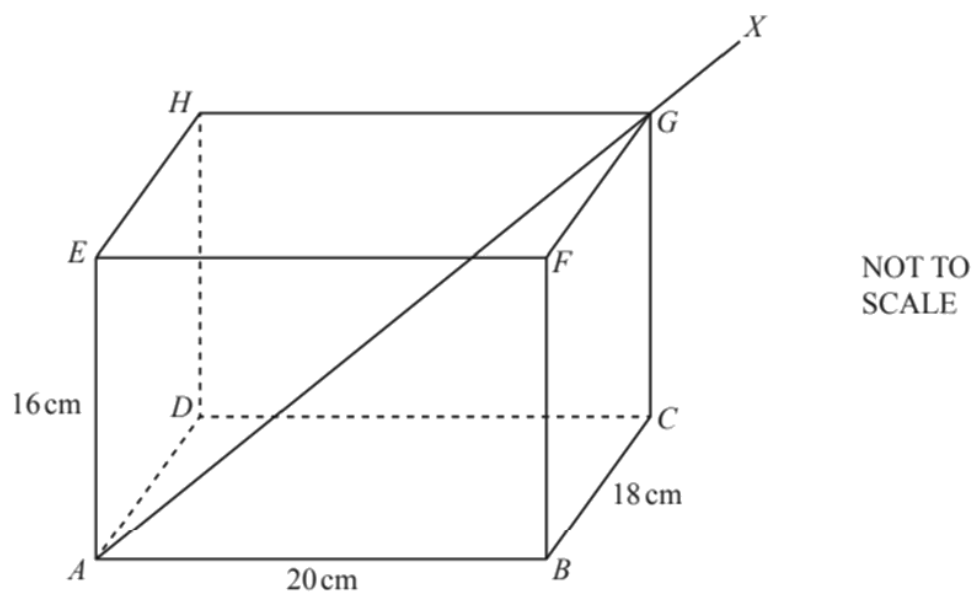
Show that $GM = 4.8$ cm, correct to 1 decimal place.

[3]

130 (a) $AC = \sqrt{18.6^2 + 9^2} = 20.66$; $\tan \theta = \frac{14.5}{20.66} \Rightarrow \boxed{35.1^\circ}$

(b) $AG = \sqrt{20.66^2 + 14.5^2} = 25.24$; $GM = 30 - 25.24 = 4.76 = 4.8 \text{ cm } \checkmark$

131 Questions



The diagram shows an open box $ABCDEFGH$ in the shape of a cuboid.

$AB = 20$ cm, $BC = 18$ cm and $AE = 16$ cm.

A thin rod AGX rests partly in the box as shown.

The rod is 40 cm long.

- (i) Calculate GX , the length of the rod which is outside the box.

$GX = \dots\dots\dots$ cm [4]

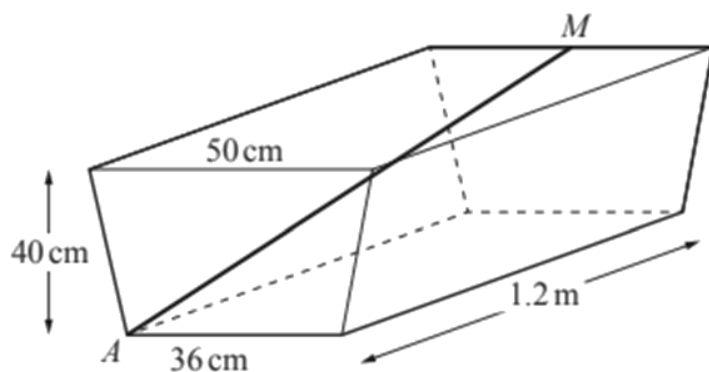
- (ii) Calculate the angle the rod makes with the base of the box.

$\dots\dots\dots$ [3]

131 (i) $AG = \sqrt{20^2 + 18^2 + 16^2} = \sqrt{980} = 31.30$; $GX = 40 - 31.30 = \boxed{8.70 \text{ cm}}$

(ii) $\sin \theta = \frac{16}{31.30} \Rightarrow \boxed{30.7^\circ}$

132

NOT TO
SCALE

A steel rod AM is placed inside the empty water trough as shown in the diagram. A is a vertex at the base of the isosceles trapezium and M is the midpoint of the top edge on the opposite face.

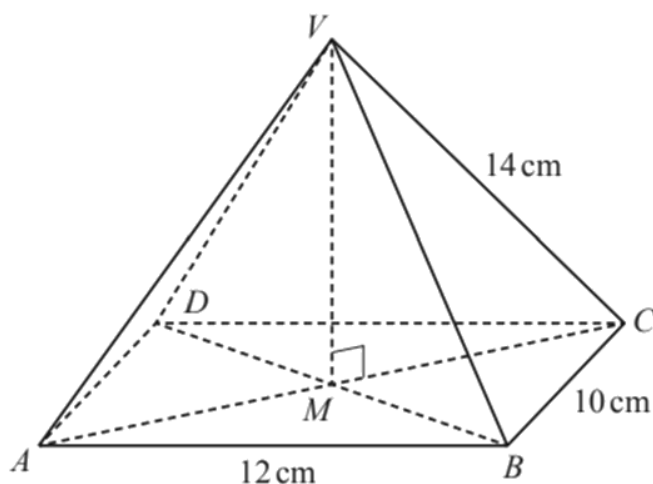
Calculate the length of the steel rod, AM .

$AM = \dots\dots\dots$ cm [4]

132 Horizontal offset across the trapezium $= \frac{36}{2} = 18$ cm (from corner to centre line), height 40, length 1.2 m = 120 cm.

$$AM = \sqrt{18^2 + 40^2 + 120^2} = \sqrt{16324} = \boxed{128 \text{ cm}}$$

133

NOT TO
SCALE

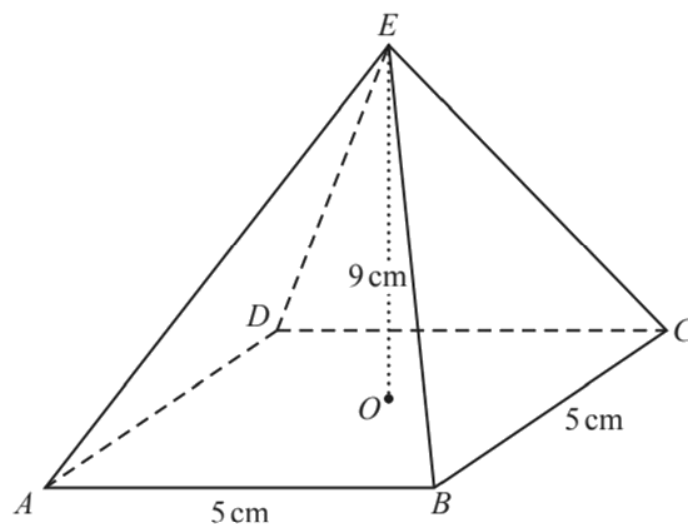
The diagram shows a pyramid $VABCD$ with a rectangular base.
 V is vertically above M , the intersection of the diagonals AC and BD .
 $AB = 12$ cm, $BC = 10$ cm and $VC = 14$ cm.

Calculate the angle that VC makes with the base $ABCD$.

..... [4]

133 $AC = \sqrt{12^2 + 10^2} = 15.62$, $MC = 7.810$. $\cos \theta = \frac{7.810}{14} \Rightarrow \boxed{56.1^\circ}$

134

NOT TO
SCALE

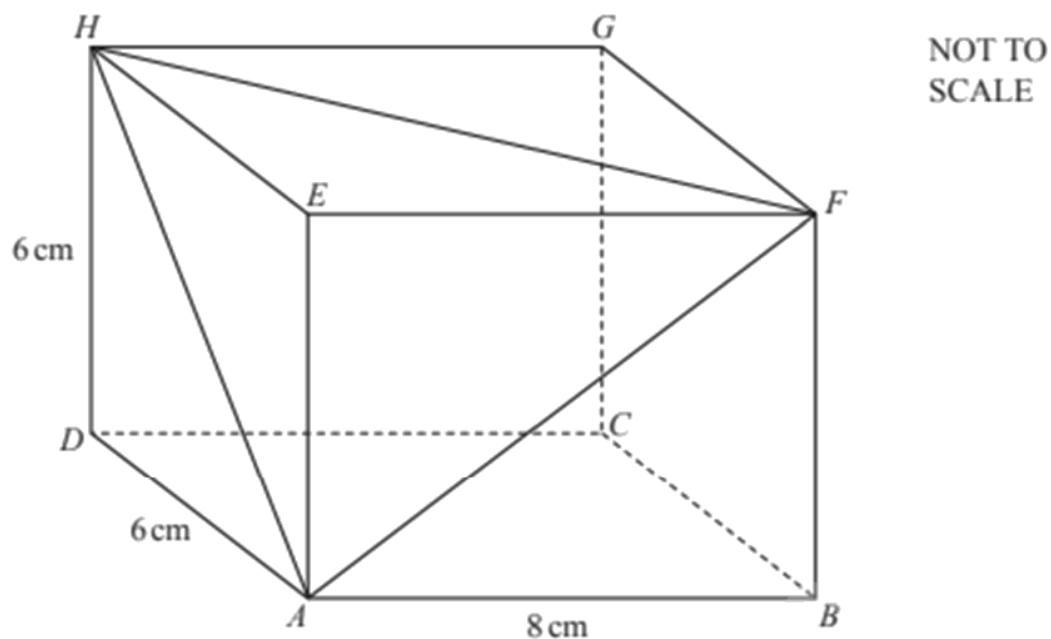
The diagram shows a pyramid $ABCDE$.
The pyramid has a square horizontal base $ABCD$ with side 5 cm.
The vertex E is vertically above the centre O of the base.
The height OE of the pyramid is 9 cm.

Calculate the angle that EC makes with the base $ABCD$.

.....

134 $AC = \sqrt{5^2 + 5^2} = 7.071$, $OC = 3.536$. $\tan \theta = \frac{9}{3.536} \Rightarrow \boxed{68.6^\circ}$

135



The diagram shows a cuboid.

$AB = 8\text{ cm}$, $AD = 6\text{ cm}$ and $DH = 6\text{ cm}$.

Calculate angle HAF .

Angle $HAF = \dots\dots\dots$ [6]

$$\begin{aligned} \mathbf{135} \quad HA &= \sqrt{6^2 + 6^2} = \sqrt{72}, \quad AF = \sqrt{8^2 + 6^2} = 10, \quad HF = \sqrt{8^2 + 6^2} = 10 \\ \cos HAF &= \frac{72 + 100 - 100}{2\sqrt{72}(10)} = 0.4243 \Rightarrow \boxed{64.9^\circ} \end{aligned}$$